

Output Without Workers: Automation and Structural Transformation

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Abstract

[Buera and Kaboski \(2009\)](#) pose a puzzle in structural transformation: why do sectoral value-added and employment shares often diverge? In this paper, value-added shares are generated by the price and income effects, while employment shares require a third force: automation's displacement effect. Inverting the growth accounting of [Jones and Tonetti \(2026\)](#), I quantify this displacement effect from national-accounts data alone. In the US, displacement accounts for 64% of the model-predicted agriculture–manufacturing and 35% of the services–manufacturing employment reallocation over 1951–1989, rising to 188% and 73% over 1990–2019. After 1990, manufacturing overtakes agriculture as the sector with the fastest automation displacement.

Keywords: structural transformation, automation, labor displacement, growth accounting.

JEL codes: E24, J24, O14, O33, O41.

1 Introduction

Structural transformation is the reallocation of economic activity across sectors. Over this process, both value-added (VA) and employment (EMP) shares decline in agriculture, are hump-shaped in manufacturing, and rise in services ([Herrendorf et al., 2014](#)). The

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structural transformation literature explains these facts through two forces: the price effect, arising from differential productivity growth combined with gross complementarity across sectoral goods (Baumol, 1967; Ngai and Pissarides, 2007), and the income effect, arising from non-homotheticity in preferences as households grow richer (Kongsamut et al., 2001).

Although sectoral VA and EMP shares move in the same direction in the long run, the magnitudes of their movements often diverge. Buera and Kaboski (2009) document this divergence (Figure 1, left) and show that, under a Cobb–Douglas production function, reconciling the two share series requires sector-specific labor distortions (Figure 1, right). The authors offer two explanations, mobility barriers and sectoral human-capital differences. Still, the required distortions are large, and what drives the VA–EMP divergence remains unresolved.

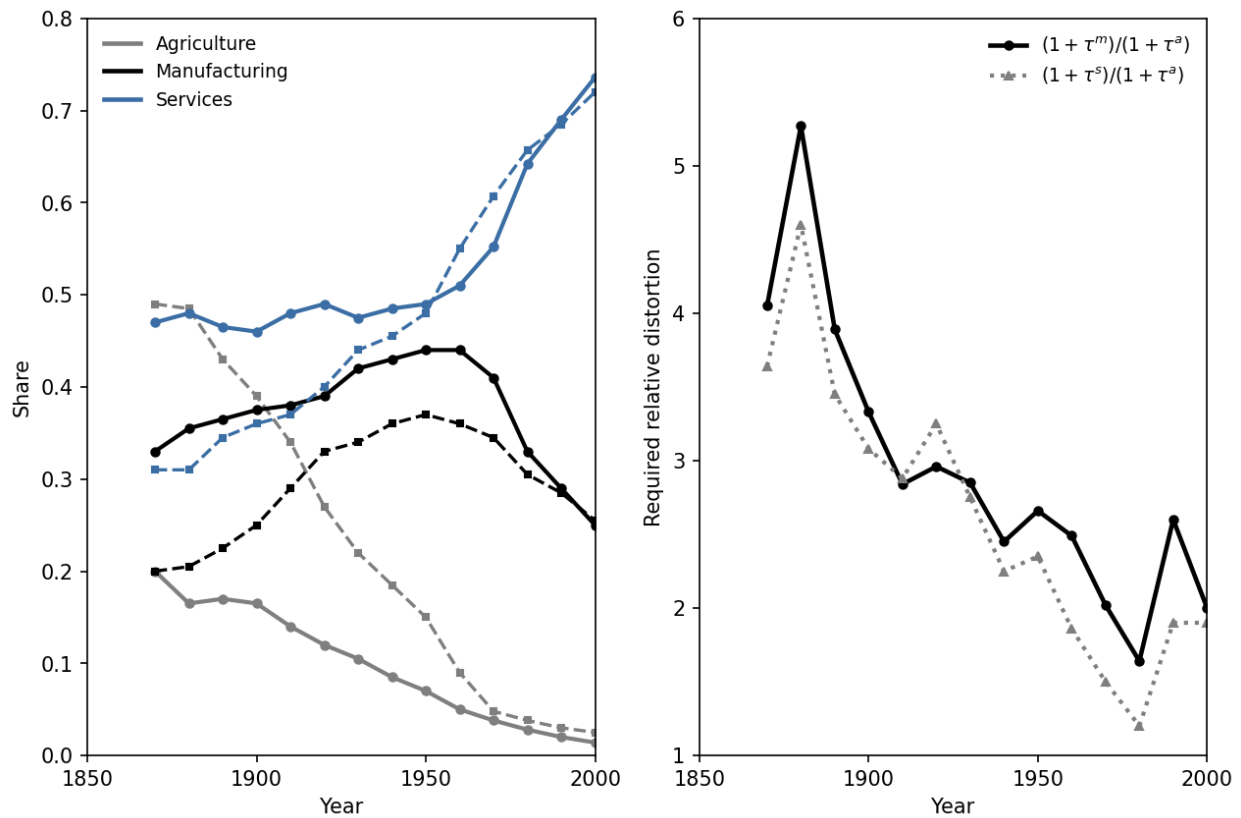


Figure 1: Left: value-added shares (solid) vs. employment shares (dashed) for agriculture (low), manufacturing (middle), and services (high); Right: implied relative sectoral labor distortions $(1 + \tau_l^m)/(1 + \tau_l^a)$ (solid) and $(1 + \tau_l^s)/(1 + \tau_l^a)$ (dotted). United States, 1870–2000. Replicated from Figure 2 of Buera and Kaboski (2009).

Where the value-added share grows faster than the employment share, sectors must be producing output without proportionate labor input: output without workers. The labor

no longer needed there must flow elsewhere, into sectors where the employment share rises faster than the value-added share. This points to a potential explanation: a technology that displaces labor at unequal rates across sectors. A growing body of literature documents labor-displacing automation at the sectoral level. In agriculture, [Manuelli and Seshadri \(2014\)](#) study the tractor’s replacement of the labor-intensive horse technology from 1910 to 1960, a period in which US farm employment fell from 11.67 million to 5.97 million full-time equivalents. In manufacturing, [Acemoglu and Restrepo \(2020\)](#) document that the stock of industrial robots in the US rose from 0.4 to 1.4 per thousand workers between 1993 and 2007, and estimate that each additional robot per thousand workers reduces the aggregate employment-to-population ratio by roughly 0.2 percentage points. Services, by contrast, are characterized by a large share of tasks (food preparation, personal care, cleaning, and child care) that, as [Autor and Dorn \(2013\)](#) argue, have so far resisted automation because they require physical dexterity and flexible interpersonal communication. They document that the share of US labor hours in such service occupations rose by about 30% between 1980 and 2005, absorbing workers displaced from routine tasks elsewhere. Taken together, this evidence suggests that automation has been a quantitatively important force reshaping sectoral labor allocation over the past century. When a machine takes over a task previously done by a worker, the sector still produces the same output but with fewer workers. Automation therefore moves employment shares without directly affecting value-added shares, which I propose as the central mechanism behind the VA–EMP divergence.

I formalize this idea by bringing a task-based production function ([Zeira, 1998](#); [Acemoglu and Restrepo, 2018, 2022](#)) into a structural-transformation setting, complementing the non-homothetic CES preferences ([Comin et al., 2021](#)) that have served as the workhorse demand-side specification. The model delivers distinct VA and EMP dynamics. Value-added share dynamics involve two forces: the price effect and the income effect. Employment share dynamics involve a third, the supply-side automation displacement effect. The displacement effect captures how automation replaces labor with capital within a sector, directly reducing sectoral employment. Because capital and labor are equally costly at the marginal task (the task at the automation frontier), this break-even condition cleanly separates the price and income effects from automation’s displacement effect. To quantify automation’s displacement effect, I build a growth-accounting strategy on [Jones and Tonetti \(2026\)](#). They measure automation (the horizontal margin) directly using large language models and recover the TFP components (the vertical margin) as a residual. I instead fix the vertical labor margin, imposing time-invariant task-level labor productivity, and recover the horizontal margin, automation’s displacement, from national accounts, requiring no task-level data.

Two clarifications about the object recovered from growth accounting are in order. First, the labor-saving term of the employment decomposition is measured directly from sectoral labor productivity and factor shares. The time-invariance of task-level labor productivity is required to interpret this term as automation's displacement, and the paper disciplines that interpretation with evidence from outside the accounting system: the estimates of aggregate task-level labor productivity growth in [Jones and Tonetti \(2026\)](#) and the task-displacement measurement of [Acemoglu and Restrepo \(2022\)](#). Second, the task space is fixed: the task set may in reality expand or shrink, notably through new task creation ([Acemoglu and Restrepo, 2019](#)), and the model abstracts from this margin.

Applying this strategy to US data over 1950–2019, I reach three main empirical conclusions. First, in the pre-1990 period, automation's displacement effect accounts for 64% of the model-predicted agriculture–manufacturing and 35% of the services–manufacturing reallocation. Second, post-1990 the displacement effect becomes the largest of the three model channels, accounting for 188% of the model-predicted agriculture–manufacturing and 73% of the services–manufacturing reallocation. The share exceeds 100% in the agriculture–manufacturing decomposition because the price and income effects, taken in isolation, would predict that agriculture's employment share should decline faster than manufacturing's. This is what the data show for value added, but the opposite holds for employment, where manufacturing declines faster. The displacement effect must therefore offset both traditional channels. Third, after 1990 manufacturing overtakes agriculture as the sector with the fastest displacement effect. This pattern is robust across alternative calibrations of the task-substitution elasticity. Decomposing the post-1950 changes in the employment shares themselves into the three channels shows that displacement directly accounts for a cumulative 6.9 percentage points of manufacturing's 21.6 percentage-point decline in employment share, 1.6 of agriculture's 11.0-point decline, and 8.4 of services' 32.6-point rise, holding the other channels at their observed values.

My framework absorbs a sizable portion of the wedge that [Buera and Kaboski \(2009\)](#) document, leaving a smaller and economically interpretable residual. The residual wedge tracks the sectoral wage gap between agriculture and the rest of the economy, and gradually declines over my 1950–2019 sample. I read this decline as a gradual erosion of the forces behind the gap, discussed in the paper.

I test the recovered labor displacement against independent micro data for the US. Because no dataset records employment at the task level, I turn to occupation-level employment from IPUMS-USA ([Ruggles et al., 2024](#)) and measure the rate at which each sector sheds workers from its automatable, declining occupations. This occupation-based measure co-moves positively with the recovered displacement (correlation 0.72 across the 21 sector–

decade cells), so the timing of the recovered measure is consistent with the micro data.

Because the method requires only standard national-accounts series, its data requirements travel far beyond the US. The policy stake is sharpest in developing economies undergoing premature deindustrialization, where manufacturing employment shares decline at lower levels of income than in earlier industrializers (Rodrik, 2016): whether those shares are falling because automation displaces workers, because demand shifts away, or because sectoral distortions persist calls for different industrial-policy responses, and the paper’s accounting separates these using data these economies already have.

The rest of the paper proceeds as follows. Section 2 develops the model and derives the value-added and employment dynamics. Section 3 lays out the measurement strategy. Section 4 reports the data sources and the technology measurements, and validates the recovered displacement against micro data. Section 5 decomposes relative employment growth into the price, income, and displacement channels plus a residual, and traces each channel’s cumulative contribution to the post-1950 evolution of the sectoral employment shares. Section 6 compares this paper’s unexplained wedge to the required distortion of Buera and Kaboski (2009). Section 7 examines the robustness of the results to alternative calibrations of the task-substitution parameter and the demand-side parameters. Section 8 concludes.

2 Model

2.1 Demand Side

I assume a continuum of measure L_t of identical households, each supplying one unit of labor inelastically, so the aggregate labor endowment equals the population L_t . Each household has non-homothetic CES preferences (Comin et al., 2021) over three sectoral goods: agriculture (a), manufacturing (m), and services (s). The consumption aggregator c_t is implicitly given by:

$$\sum_{j \in \{a, m, s\}} \Omega_j^{\frac{1}{\sigma}} \left(\frac{c_t^j}{\varepsilon_j} \right)^{\frac{\sigma-1}{\sigma}} = 1, \quad (1)$$

where $\Omega_j > 0$ are taste weights. The parameter $\sigma \in (0, 1)$ governs the elasticity of substitution across sectoral goods. Because $\sigma < 1$, sectoral goods are gross complements, so that the expenditure share rises for goods with rising relative prices. Parameters $\varepsilon_j > 0$ control non-homotheticity: as the consumption aggregator c_t grows, expenditure shifts

toward goods with higher ε_j . Later in the paper I calibrate $\varepsilon_s > \varepsilon_m > \varepsilon_a$, consistent with the empirical pattern that services are the most income-elastic and agriculture the least.

For tractability, I abstract from household saving and intertemporal optimization. Each household consumes its entire wage income within the period, so the budget constraint is

$$\sum_{j \in \{a, m, s\}} P_t^j c_t^j = W_t. \quad (2)$$

Marshallian demand yields

$$c_t^j = \Omega_j \left(\frac{P_t^j}{W_t} \right)^{-\sigma} c_t^{(1-\sigma)\varepsilon_j}. \quad (3)$$

The corresponding sectoral expenditure is

$$P_t^j c_t^j = \Omega_j (P_t^j)^{1-\sigma} W_t^\sigma c_t^{(1-\sigma)\varepsilon_j}. \quad (4)$$

2.2 Supply Side

Production uses two factors: labor, supplied inelastically by households, and capital, produced by a separate sector k . I treat capital as an intermediate good that fully depreciates after use. This assumption allows me to abstract from dynamic optimization over capital accumulation and focus on the within-period allocation of tasks between factors. Separating capital production into its own sector also ensures that the value-added shares of agriculture, manufacturing, and services equal their consumption expenditure shares.

All markets are perfectly competitive: in each sector a representative firm takes the sectoral output price, the capital-good price, and the wage as given. Because the technology below exhibits constant returns to scale, equilibrium profits are zero and each sectoral good price equals the minimized unit cost of production.

For each sector $j \in \{k, a, m, s\}$, output Y_t^j is produced by aggregating a unit continuum of task-level outputs $y_t^j(i)$ ([Acemoglu and Restrepo, 2022](#)):

$$Y_t^j = \left[\int_0^1 y_t^j(i)^\rho di \right]^{1/\rho}, \quad (5)$$

where $\rho < 0$ implies gross complementarity among tasks. Each task can be produced by

either capital or labor, which are perfect substitutes at the task level:

$$y_t^j(i) = \psi_t^{k,j}(i) k_t^j(i) + \psi_t^{l,j}(i) l_t^j(i), \quad (6)$$

where $\psi_t^{k,j}(i)$ is the productivity of capital and $\psi_t^{l,j}(i)$ is the productivity of labor in task i ; labor productivity carries no time subscript because I maintain that it is time-invariant, a restriction I motivate below.

Since capital and labor are perfect substitutes within each task, cost minimization implies that firms use whichever factor is cheaper. The unit cost of task i in sector j is

$$p_t^j(i) = \begin{cases} P_t^k / \psi_t^{k,j}(i), & \text{if produced by capital,} \\ W_t / \psi_t^{l,j}(i), & \text{if produced by labor,} \end{cases} \quad (7)$$

where W_t is the wage and P_t^j denotes the price of sector j 's output, so that P_t^k is the price of the capital good produced by sector k .

I rank tasks so that capital's comparative advantage $\psi_t^{k,j}(i) / \psi_t^{l,j}(i)$ is decreasing in i , and assume that this ratio is continuous in i and that there is a task $\beta_t^j \in (0, 1)$ at which capital and labor are equally costly, the break-even task:

$$\frac{P_t^k}{\psi_t^{k,j}(\beta_t^j)} = \frac{W_t}{\psi_t^{l,j}(\beta_t^j)}. \quad (8)$$

Thus, tasks in $[0, \beta_t^j]$ are produced by capital and tasks in $[\beta_t^j, 1]$ by labor. An expansion of β_t^j (the process of automation) reflects either a rise in capital productivity at the marginal task or a fall in the relative cost of capital to labor. By the break-even condition (8), movements in β_t^j reshape factor allocation without altering marginal cost.

Cost minimization yields task-level factor demands. For tasks in $[0, \beta_t^j]$, capital demand is

$$k_t^j(i) = \left(P_t^k\right)^{-\frac{1}{1-\rho}} \psi_t^{k,j}(i)^{\frac{\rho}{1-\rho}} \left(P_t^j\right)^{\frac{1}{1-\rho}} Y_t^j, \quad (9)$$

and for tasks in $[\beta_t^j, 1]$, labor demand is

$$l_t^j(i) = W_t^{-\frac{1}{1-\rho}} \psi_t^{l,j}(i)^{\frac{\rho}{1-\rho}} \left(P_t^j\right)^{\frac{1}{1-\rho}} Y_t^j. \quad (10)$$

Setting each sectoral price equal to the minimized unit cost yields the sectoral price index:

$$P_t^j = \left[\int_0^{\beta_t^j} \left(\frac{P_t^k}{\psi_t^{k,j}(i)} \right)^{-\frac{\rho}{1-\rho}} di + \int_{\beta_t^j}^1 \left(\frac{W_t}{\psi_t^{l,j}(i)} \right)^{-\frac{\rho}{1-\rho}} di \right]^{-\frac{1-\rho}{\rho}}. \quad (11)$$

2.3 Relative Value Added and Employment Dynamics

The demand and supply blocks above are closed by market clearing. I now define the equilibrium.

Definition 1 (Competitive equilibrium). Given task-level productivities $\{\psi_t^{k,j}(\cdot), \psi_t^{l,j}(\cdot)\}_{j \in \{k,a,m,s\}}$ and the labor endowment L_t , a competitive equilibrium at date t is a collection of prices $\{P_t^j\}_{j \in \{k,a,m,s\}}$ and a wage W_t , together with an allocation, consisting of consumption $\{c_t^j\}_{j \in \{a,m,s\}}$ and task-level factor inputs $\{k_t^j(i), l_t^j(i)\}_{i \in [0,1]}$, with sectoral aggregates $K_t^j \equiv \int_0^1 k_t^j(i) di$ and $L_t^j \equiv \int_0^1 l_t^j(i) di$ and sectoral outputs Y_t^j given by the technology (5)–(6), such that:

- (i) (*Household optimization*) given prices and the wage, $\{c_t^j\}$ maximizes the consumption aggregator (1) subject to the budget constraint (2);
- (ii) (*Firm optimization*) in every sector, firms choose the quantities of capital and labor at each task to minimize the cost of production given prices, and competition sets each sectoral price equal to the minimized unit cost;
- (iii) (*Market clearing*) labor, $\sum_{j \in \{k,a,m,s\}} L_t^j = L_t$; the capital good, $\sum_{j \in \{k,a,m,s\}} K_t^j = Y_t^k$; and each consumption good, $L_t c_t^j = Y_t^j$ for $j \in \{a, m, s\}$.

Definition 1 holds period by period: as the task-level technologies evolve, the equilibrium prices, allocations, and automation frontiers trace out the sectoral dynamics that the rest of the paper characterizes.

The model accommodates two conceptually distinct dimensions of technical change. The first is vertical: capital productivity growth at tasks $\hat{\psi}_t^{k,j}(i)$ ¹. The second is horizontal: the expansion of the automation frontier, $\hat{\beta}_t^j$, which reassigns tasks from labor to capital. A key feature of the model is that these two dimensions enter sectoral aggregates through distinct channels: the break-even condition at the automation frontier cleanly separates their effects, as I show below.

I maintain that task-level labor productivity $\psi_t^{l,j}(i)$ is time-invariant, shutting down the

¹Throughout, a dot denotes a derivative with respect to time, $\dot{x}_t \equiv \partial x_t / \partial t$, and a hat denotes a proportional growth rate $\hat{x}_t \equiv \dot{x}_t / x_t$.

labor-side vertical margin. The restriction is dictated by the accounting: allowing $\psi^{l,j}(i)$ to vary over time introduces a fourth technology object into a system that, as Section 3 shows, delivers only three independent measurement conditions per sector, so one object must be fixed to recover the rest. I fix this margin for two further reasons. First, the available evidence indicates it is small: [Jones and Tonetti \(2026\)](#) estimate its counterpart to be small for the postwar US, in the private business sector and in agriculture. Second, doing so yields a method requiring only standard national accounts, which is what allows it to travel to any economy with KLEMS-style data.

I next introduce notation for task shares of sectoral factor use and a task-share-weighted average of sectoral capital productivity growth. Let $\omega_t^{k,j}(i)$ and $\omega_t^{l,j}(i)$ denote task i 's share of sectoral capital use and labor use, respectively:

$$\omega_t^{k,j}(i) \equiv \frac{k_t^j(i)}{K_t^j} = \frac{\psi_t^{k,j}(i)^{\frac{\rho}{1-\rho}}}{\int_0^{\beta_t^j} \psi_t^{k,j}(i')^{\frac{\rho}{1-\rho}} di'}, \quad i \in [0, \beta_t^j], \quad (12)$$

$$\omega_t^{l,j}(i) \equiv \frac{l_t^j(i)}{L_t^j} = \frac{\psi_t^{l,j}(i)^{\frac{\rho}{1-\rho}}}{\int_{\beta_t^j}^1 \psi_t^{l,j}(i')^{\frac{\rho}{1-\rho}} di'}, \quad i \in [\beta_t^j, 1]. \quad (13)$$

The second equality in each of (12)–(13) follows from the task-level factor demands (9)–(10). I then define $\hat{\psi}_t^{k,j}$, the task-share-weighted average of sectoral capital productivity growth:

$$\hat{\psi}_t^{k,j} \equiv \int_0^{\beta_t^j} \omega_t^{k,j}(i) \hat{\psi}_t^{k,j}(i) di. \quad (14)$$

This is capital productivity growth on already-automated tasks: it collapses the heterogeneous task-level components $\{\hat{\psi}_t^{k,j}(i)\}_{i \in [0, \beta_t^j]}$ into a single sectoral object that summarizes the vertical dimension of technical change in sector j .

Combining the Marshallian demand (3), the task-level factor demands (9)–(10), and goods-market clearing yields the two central equations of the paper, characterizing relative sectoral value-added and employment dynamics.

Proposition 1 (Sectoral value-added and employment dynamics). *In the competitive equilibrium of Definition 1, for any pair of sectors $j, j' \in \{a, m, s\}$, relative value-added dynamics follow*

$$\widehat{VA}_t^j - \widehat{VA}_t^{j'} = \underbrace{(1 - \sigma) \left(\hat{P}_t^j - \hat{P}_t^{j'} \right)}_{\text{Price effect}} + \underbrace{(1 - \sigma)(\varepsilon_j - \varepsilon_{j'})}_{\text{Income effect}} \hat{c}_t, \quad (15)$$

and relative employment dynamics follow

$$\hat{L}_t^j - \hat{L}_t^{j'} = \underbrace{\left(1 - \sigma + \frac{\rho}{1-\rho}\right) \left(\hat{P}_t^j - \hat{P}_t^{j'}\right)}_{\text{Price effect}} + \underbrace{(1 - \sigma)(\varepsilon_j - \varepsilon_{j'}) \hat{c}_t}_{\text{Income effect}} - \underbrace{\left[\omega_t^{l,j}(\beta_t^j) \dot{\beta}_t^j - \omega_t^{l,j'}(\beta_t^{j'}) \dot{\beta}_t^{j'}\right]}_{\text{Automation's displacement effect}}. \quad (16)$$

Proof. See Appendix A.3. □

Consider (15) first. Given sectoral prices P_t^j and the wage W_t , households choose sectoral consumption shares according to the non-homothetic CES preferences. Goods market clearing then equates sectoral value added to sectoral consumption expenditure for each of the three sectors, as the capital-producing sector k absorbs all investment activity. As a consequence, relative value-added dynamics are driven entirely by the price and income effects of (15). The VA share tilts toward the sector with rising relative prices and larger income elasticity as specified in the non-homothetic CES preferences.

Relative employment dynamics (16) are a natural consequence of aggregating task-level labor demand across the sector. Intuitively, when firms decide how much labor to hire, they respond to quantities of goods demanded, sectoral prices, wage, and technologies. Quantities demanded are governed by the demand-side price and income effects in the Marshallian demand (3): a rise in the relative price of sector j 's good lowers the quantity households demand, while an expansion of aggregate consumption shifts demand toward income-elastic sectors. The rising price of the sectoral good simultaneously raises revenue per unit sold and thus encourages more production, raising labor demand. These two price responses offset each other, and their net contribution to relative employment dynamics is captured by the price-effect coefficient $1 - \sigma + \rho/(1 - \rho)$, which is small under the calibration in Section 5. The third term, the difference in the sectoral displacement rates $\omega_t^{l,j}(\beta_t^j) \dot{\beta}_t^j$, labeled automation's displacement effect, is unique to employment dynamics. Since $\omega_t^{l,j}(\beta_t^j) = l_t^j(\beta_t^j)/L_t^j$ is the marginal task's share of sectoral labor at the automation frontier β_t^j , multiplying it by the rate at which the frontier advances, $\dot{\beta}_t^j$, gives precisely the rate at which sectoral labor is displaced by newly automated tasks.

One might ask whether automation, as a form of technological advance, should not enter both the price effect and the income effect in (15)–(16). The model's answer is no, at least not directly.

Lemma 1 (First-order invariance of prices and consumption to the automation frontier's movement). *In the competitive equilibrium of Definition 1, all prices and the consumption*

aggregator are first-order invariant to the automation frontier taking the wage as numeraire,² for all $j, j' \in \{k, a, m, s\}$,

$$\frac{d \left(P_t^j / W_t \right)}{d \beta_t^{j'}} = 0 \quad \text{and} \quad \frac{dc_t}{d \beta_t^{j'}} = 0.$$

Proof. See Appendix A.4. □

Intuitively, at the moment automation occurs, the newly automated tasks lie exactly at the break-even frontier, where the unit cost of producing with capital equals the unit cost of producing with labor. Reassigning a marginal task across factors therefore leaves the sectoral cost, and hence the sectoral price, unchanged at first order. This can be seen directly in the sectoral price growth rate. Log-differentiating the price index (11) and applying the break-even condition yields

$$\hat{P}_t^j = \alpha_t^j \hat{P}_t^k + (1 - \alpha_t^j) \hat{W}_t - \alpha_t^j \hat{\psi}_t^{k,j}, \quad (17)$$

where $\alpha_t^j \equiv P_t^k K_t^j / (P_t^j Y_t^j)$ is sector j 's capital cost share. The automation frontier change $\dot{\beta}_t^j$ does not appear. Sectoral price growth is thus driven entirely by the vertical margins, as the two terms that remain, wage growth $\hat{W}_t$ and capital-good price growth $\hat{P}_t^k$, are themselves driven by the vertical margin. To see this, take the wage as numeraire, $\hat{W}_t = 0$: capital's own price growth equation, (17) for $j = k$, then ties capital-good price growth to its vertical margin, $\hat{P}_t^k = -\frac{\alpha_t^k}{1 - \alpha_t^k} \hat{\psi}_t^{k,k}$, and substituting back into (17) collapses every sectoral price growth rate to a combination of vertical margins,

$$\hat{P}_t^j = -\alpha_t^j \left(\frac{\alpha_t^k}{1 - \alpha_t^k} \hat{\psi}_t^{k,k} + \hat{\psi}_t^{k,j} \right).$$

Consumption growth $\hat{c}_t$, in turn, is also driven by the vertical margins alone: c_t is determined by the household's problem given the wage and sectoral prices, whose growth rates are themselves functions of the vertical margins.

However, cumulative automation does shape sectoral prices indirectly. As the frontier advances, it shifts the fraction of tasks produced by capital, changing the factor share α_t^j and the set of capital tasks entering $\hat{\psi}_t^{k,j}$. In the model, therefore, automation operates on the price and income effects in the long run, but its direct impact, at first order, is only to

²The result does not depend on the choice of numeraire: because every relative price P_t^j / W_t is invariant, so is any price ratio $P_t^j / P_t^{j'}$, and the statement holds taking any price as the numeraire.

displace labor.

3 Measurement Strategy

The empirical question that follows is how quantitatively important the sectoral displacement rate $\omega_t^{l,j}(\beta_t^j) \dot{\beta}_t^j$ is for cross-sector employment reallocation. Answering it requires a measurement strategy, because no dataset records which task sits at a sector's automation frontier β_t^j or how much labor has been displaced from that task.

I therefore back the displacement out of the macro aggregates that any KLEMS-style sectoral account records: real value added, employment, and labor cost share. The strategy builds on the growth-accounting approach of [Jones and Tonetti \(2026\)](#), inverting their logic: where they measure the horizontal margin directly and solve for the vertical, I fix the vertical labor margin and solve for the horizontal. The output of the procedure is three sectoral technology objects. The vertical object is capital-augmenting growth at already-automated tasks; the two horizontal objects are the automation frontier's contributions to capital deepening and to labor displacement. The analysis in this paper draws only on the displacement.

The strategy proceeds in three steps: aggregate the task-based production function into a sectoral CES representation; recover the CES technology aggregates from macro data; and decompose them into the three task-based objects.

3.1 CES Representation

The task-based production function admits a CES representation, derived in Appendix A.5. Define the sectoral capital and labor productivity aggregates

$$B_t^j \equiv \left[\int_0^{\beta_t^j} \psi_t^{k,j}(i)^{\frac{\rho}{1-\rho}} di \right]^{\frac{1-\rho}{\rho}}, \quad (18)$$

$$A_t^j \equiv \left[\int_{\beta_t^j}^1 \psi_t^{l,j}(i)^{\frac{\rho}{1-\rho}} di \right]^{\frac{1-\rho}{\rho}}. \quad (19)$$

Sectoral output can then be written as

$$Y_t^j = \left[\left(B_t^j K_t^j \right)^\rho + \left(A_t^j L_t^j \right)^\rho \right]^{1/\rho}. \quad (20)$$

This representation maps the task-based objects onto two sectoral technology aggregates.

A_t^j and B_t^j each depend on β_t^j through the integration domains in (18)–(19), so their growth rates move with $\dot{\beta}_t^j$. Under the maintained assumption that task-level labor productivity $\psi^{l,j}(i)$ is time-invariant, $\hat{A}_t^j$ is driven entirely by movements of the automation frontier, while $\hat{B}_t^j$ combines movements of the frontier with task-level capital-augmenting growth.

3.2 Recovering A and B from Macro Data

A_t^j and B_t^j in the CES form can be measured from macro aggregates. From the first-order condition for labor (Appendix A.6),

$$A_t^j = \left(1 - \alpha_t^j\right)^{1/\rho} \frac{Y_t^j}{L_t^j}, \quad (21)$$

which expresses A_t^j as a function of sectoral output, employment, and the labor cost share. $\hat{B}_t^j$ then follows from the standard TFP accounting identity, where $\widehat{\text{TFP}}_t^j$ is the sectoral Solow residual (Appendix A.7),

$$\widehat{\text{TFP}}_t^j = \alpha_t^j \hat{B}_t^j + \left(1 - \alpha_t^j\right) \hat{A}_t^j, \quad (22)$$

which can be rearranged to

$$\hat{B}_t^j = \frac{\widehat{\text{TFP}}_t^j - \left(1 - \alpha_t^j\right) \hat{A}_t^j}{\alpha_t^j}. \quad (23)$$

Both $\hat{A}_t^j$ and $\hat{B}_t^j$ are therefore recovered from four standard sectoral series: output, employment, the labor cost share, and total factor productivity.

3.3 Mapping to Task-Based Objects

With A_t^j and B_t^j recovered, I decompose them into the underlying task-based objects. Because $\psi^{l,j}(i)$ is time-invariant, $\hat{A}_t^j$ moves only with the automation frontier and delivers a single object, automation's labor displacement $\omega_t^{l,j}(\beta_t^j)\dot{\beta}_t^j$. The capital aggregate $\hat{B}_t^j$ moves with both the frontier and task-level capital-augmenting productivity, so it bundles two objects, automation's capital deepening $\omega_t^{k,j}(\beta_t^j)\dot{\beta}_t^j$ and capital-augmenting growth at already-automated tasks $\hat{\psi}_t^{k,j}$. With the break-even condition (8) supplying the one extra

restriction, differentiating (18)–(19) pins down all three:

$$\omega_t^{k,j}(\beta_t^j)\dot{\beta}_t^j = -\frac{\rho}{1-\rho} \frac{1-\alpha_t^j}{\alpha_t^j} \hat{A}_t^j, \quad (24)$$

$$\omega_t^{l,j}(\beta_t^j)\dot{\beta}_t^j = -\frac{\rho}{1-\rho} \hat{A}_t^j, \quad (25)$$

$$\hat{\psi}_t^{k,j} = \hat{B}_t^j + \frac{1-\alpha_t^j}{\alpha_t^j} \hat{A}_t^j. \quad (26)$$

The shares $\omega_t^{k,j}(\beta_t^j) = k_t^j(\beta_t^j)/K_t^j$ and $\omega_t^{l,j}(\beta_t^j) = l_t^j(\beta_t^j)/L_t^j$ are the fractions of sectoral capital and labor used by the marginal task. As the frontier advances at rate $\dot{\beta}_t^j$, this task switches from labor to capital, so $\omega_t^{k,j}(\beta_t^j)\dot{\beta}_t^j$ in (24) is the rate at which newly automated tasks add to sectoral capital (automation's capital deepening), while $\omega_t^{l,j}(\beta_t^j)\dot{\beta}_t^j$ in (25) is the rate at which they remove labor from the sector (automation's labor displacement). Equation (26) captures capital-augmenting technical change, the productivity gain $\hat{\psi}_t^{k,j}$ from continued advancement in tasks already automated.

The object of interest is automation's labor displacement $\omega_t^{l,j}(\beta_t^j)\dot{\beta}_t^j$. Taking growth rates of (21) and substituting into (25) expresses it in observables:

$$\omega_t^{l,j}(\beta_t^j)\dot{\beta}_t^j = -\frac{\rho}{1-\rho} (\hat{Y}_t^j - \hat{L}_t^j) - \frac{1}{1-\rho} \frac{d \ln(1-\alpha_t^j)}{dt}. \quad (27)$$

Displacement is therefore recovered from two published series alone: sectoral labor-productivity growth and labor-share growth. Sectors with rapid labor-productivity growth and declining labor shares are, through the lens of the model, sectors reassigning tasks to capital.

3.4 The Impossible Quartet

The three measurement equations (24)–(26) recover three task-based objects from three conditions: the two macro inputs $(\hat{A}_t^j, \hat{B}_t^j)$ and the break-even restriction. A natural question is whether the time-invariance restriction could be dropped, with the data determining the labor-side vertical margin as well. The following proposition shows that it cannot: a fourth object exceeds what three conditions can determine.

Proposition 2 (The impossible quartet). *Suppose sectoral output is produced by the task-based technology of Section 2 but with varying task-level labor productivities, so that output admits the CES representation (20) with sectoral productivity aggregates (A_t^j, B_t^j) . Then their growth rates*

$(\hat{A}_t^j, \hat{B}_t^j)$, together with the break-even condition (8), provide exactly three measurement conditions per sector on the four technology objects $\{\omega_t^{l,j}(\beta_t^j)\dot{\beta}_t^j, \omega_t^{k,j}(\beta_t^j)\dot{\beta}_t^j, \hat{\psi}_t^{k,j}, \hat{\psi}_t^{l,j}\}$:

$$\begin{aligned} -\frac{1-\rho}{\rho} \omega_t^{l,j}(\beta_t^j)\dot{\beta}_t^j + \hat{\psi}_t^{l,j} &= \hat{A}_t^j, \\ \frac{1-\rho}{\rho} \omega_t^{k,j}(\beta_t^j)\dot{\beta}_t^j + \hat{\psi}_t^{k,j} &= \hat{B}_t^j, \\ \omega_t^{k,j}(\beta_t^j)\dot{\beta}_t^j - \frac{1-\alpha_t^j}{\alpha_t^j} \omega_t^{l,j}(\beta_t^j)\dot{\beta}_t^j &= 0. \end{aligned} \tag{28}$$

The four objects are therefore not jointly identified. Exact identification requires a fourth, task-level condition.

Jones and Tonetti (2026) and this paper supply the fourth condition in two different ways. Jones and Tonetti take the measurement route: using large language models to date the automation of tasks, they measure $\omega_t^{l,j}(\beta_t^j)\dot{\beta}_t^j$ directly and recover the rest. I take the restriction route, imposing $\hat{\psi}_t^{l,j} = 0$ to fix the vertical labor margin and recover the rest from the three sectoral conditions alone. This route requires only four series, recorded in any KLEMS-style account: real output, employment, the labor cost share, and total factor productivity. The method therefore travels to economies and periods without task-level data, though each application must judge whether $\hat{\psi}_t^{l,j} = 0$ remains plausible there.

Two pieces of evidence discipline this restriction. First, Jones and Tonetti (2026) estimate the counterpart of $\hat{\psi}_t^{l,j}$ in their paper at -0.5% per year for the US private business sector over 1950–2023 and $+0.4\%$ per year for agriculture over 1950–2021, windows that cover my full sample, at a task elasticity of 0.3, the closest value they report to my baseline. The implied bias follows from Proposition 2: measured displacement equals true displacement plus $-\frac{\rho}{1-\rho} \hat{\psi}_t^{l,j}$, i.e., two-thirds of $\hat{\psi}_t^{l,j}$ at $\rho = -2$. At these magnitudes the bias is roughly 0.3 percentage points per year, small relative to the recovered displacement rates of 1–3% per year (Table 1). Second, Acemoglu and Restrepo (2022) tie within-industry labor-share declines over 1987–2016 to automation-driven task displacement: their automation proxies account for 45% of observed changes in industry labor shares (largest in industries at the forefront of automation adoption), while factor-augmenting technology plays a limited role and offshoring is quantitatively secondary.

One piece of evidence cuts the other way: Jones and Tonetti’s estimate for agriculture is positive. A positive $\hat{\psi}_t^{l,a}$ inflates measured displacement, so the large agricultural displacement recovered below (Tables 1 and 3) is an upper bound. The bias, however, strengthens the central finding that manufacturing overtakes agriculture in displacement after 1990: correcting an overstated agricultural series would only widen the post-1990 manufacturing–agriculture gap. For that finding to be spurious, task-level labor productivity growth in

manufacturing would have to exceed agriculture’s by more than 1.2 percentage points per year at the baseline, the opposite of what the evidence indicates.

4 Empirical Results

4.1 Data Sources

I apply the framework to the US over 1950–2019. Measuring automation’s sectoral displacement and capital-deepening rates requires sectoral real value added, employment, and factor share data, taken from BEA industry accounts and NIPA Table 6.8. Sectoral TFP additionally identifies capital-augmenting technology at already-automated tasks. Although automation’s capital-deepening rate and capital-augmenting technical change are not central to the argument, Table 1 reports them as well to complete the accounting strategy. Detailed data sources are listed in Appendix B.

I aggregate industries into agriculture, manufacturing, and services, following [Comin et al. \(2021\)](#) (CLM hereafter). The agricultural sector consists of agriculture, forestry, fishing, and hunting. The manufacturing sector consists of mining, construction, and manufacturing. The service sector consists of all remaining private industries, including utilities but excluding government. I exclude government to remain consistent with the demand-side calibration in Section 5, which uses CLM’s Table X col. 6 estimates. Those estimates come from US Consumer Expenditure Survey (CEX) household data, recording only private household consumption.

Following [Herrendorf et al. \(2015\)](#), I reallocate the rent paid to nonoperator landlords from services to agriculture when computing value added. The capital cost share is constructed as

$$\alpha_t^j = \frac{GOS_t^j - PI_t^j}{CE_t^j + GOS_t^j - PI_t^j},$$

where CE_t^j , GOS_t^j , PI_t^j are compensation of employees, gross operating surplus, and proprietors’ income. Subtracting proprietors’ income from the capital share assigns self-employment income to labor, following the adjustment of [Gollin \(2002\)](#).

For employment I use NIPA Table 6.8 “Persons Engaged in Production by Industry,” which counts full- and part-time employees together with the self-employed.

The US shifted its industry classification from the SIC to the NAICS standard between 1997 and 1998; I drop the 1997–1998 growth rate from all growth-rate-based calculations to avoid a spurious jump. Because the model’s formulas are derived in continuous time, I

evaluate share weights as two-year averages, $\bar{\alpha}_t^j = \frac{1}{2}(\alpha_t^j + \alpha_{t-1}^j)$, the standard Törnqvist discretization of the Divisia index.

4.2 Technology Measurement

Table 1 reports the three technology objects recovered by the measurement procedure: capital-augmenting growth at already-automated tasks $\hat{\psi}_t^{k,j}$ (vertical), and the automation frontier's contributions to capital deepening $\omega_t^{k,j}(\beta_t^j)\dot{\beta}_t^j$ and to labor displacement $\omega_t^{l,j}(\beta_t^j)\dot{\beta}_t^j$ (horizontal). All entries are in percent per year at $\rho = -2$, corresponding to an elasticity of substitution across tasks $\sigma_{KL} = 1/3$, in line with the meta-analysis in [Gechert et al. \(2022\)](#).³

Table 1: Sectoral technology measurements, United States 1951–2019 ($\rho = -2$)

	$\omega^{k,j}(\beta^j)\dot{\beta}^j$			$\omega^{l,j}(\beta^j)\dot{\beta}^j$			$\hat{\psi}^{k,j}$		
	A	M	S	A	M	S	A	M	S
1950s	2.53	4.67	2.77	3.34	1.70	1.72	5.75	3.12	1.64
1960s	2.86	4.16	2.15	4.41	1.51	1.38	6.03	7.06	2.50
1970s	1.13	2.57	1.16	2.14	0.88	0.67	2.48	−0.64	1.89
1980s	0.94	4.72	0.73	2.25	1.89	0.48	6.53	4.13	−0.48
1990s	0.16	4.69	1.22	0.13	2.12	0.75	6.25	4.55	0.51
2000s	1.35	4.83	2.09	2.62	2.98	1.30	5.08	5.03	0.72
2010s	0.27	0.67	1.07	0.48	0.55	0.75	−0.05	1.22	1.28
Pre-1990	1.85	4.01	1.67	3.03	1.49	1.05	5.19	3.43	1.38
Post-1990	0.61	3.35	1.47	1.11	1.88	0.94	3.68	3.57	0.85

Notes: Values in percent per year. A = Agriculture, M = Manufacturing, S = Services. $\omega^{k,j}(\beta^j)\dot{\beta}^j$ = automation's capital deepening; $\omega^{l,j}(\beta^j)\dot{\beta}^j$ = the sectoral displacement rate; $\hat{\psi}^{k,j}$ = capital-augmenting growth at automated tasks. Decade and sub-period entries are period averages of annual values. Pre-1990 = 1951–1989; Post-1990 = 1990–2019.

The sectoral displacement rate $\omega_t^{l,j}(\beta_t^j)\dot{\beta}_t^j$ is the rate at which labor in sector j is reassigned to capital by the expansion of the automation frontier. The measurements reveal a reordering around 1990. Pre-1990, the cross-sector ranking is $A > M > S$, with agriculture displacing labor at 3.03% per year, manufacturing at 1.49%, and services at 1.05%. Post-1990, the ranking flips between agriculture and manufacturing: manufacturing rises to 1.88% per year while agriculture falls to 1.11%, with services again last at 0.94%. The natural reading is that labor-displacing automation shifted from agriculture, where it operated for most of the twentieth century via mechanization, to manufacturing, where industrial robots,

³ $\sigma_{KL} = 1/(1 - \rho)$ is the elasticity of substitution between tasks (equivalently, between capital and labor) within a sector.

programmable controllers, and computerized process automation expanded sharply from the late 1980s onward. Import competition displaced manufacturing labor over the same period (Autor et al., 2013; Pierce and Schott, 2016); within the model, tasks moved offshore are isomorphic to tasks moved to capital, so my measure bundles the two. I examine the robustness of this shift in Section 7.

The other two margins behave as follows. Capital-augmenting growth $\hat{\psi}_t^{k,j}$ is positive in nearly all sector-decade pairs, and the cross-sector ranking $\hat{\psi}^{k,a} > \hat{\psi}^{k,m} > \hat{\psi}^{k,s}$ is stable across both subperiods. Before 1990 it averages 5.19% per year in agriculture, 3.43% in manufacturing, and 1.38% in services. After 1990 all three rates fall, to 3.68% in agriculture, 3.57% in manufacturing, and 0.85% in services. Productivity at already-automated tasks has thus risen fastest in agriculture and slowest in services. The frontier’s contribution to capital deepening $\omega^{k,j}(\beta^j)\dot{\beta}^j$ is largest in manufacturing, where it averages 4.01% per year before 1990 and 3.35% after. In services it is more moderate, averaging 1.67% before 1990 and 1.47% after. In agriculture it falls sharply over time, from 1.85% before 1990 to 0.61% after.

4.3 Validating the Displacement Measure with Micro Data

The displacement $\omega_t^{l,j}(\beta_t^j)\dot{\beta}_t^j$ in Table 1 is recovered from macro data on sectoral value added, employment, and factor shares; this section validates it against micro data. The recovered object is task-level, and no dataset records employment at the task level, so the displacement cannot be measured directly. Occupation-level data from IPUMS-USA offer the closest available proxy. As the automation frontier advances, the occupations that performed the newly automated tasks should be shed from sector j . I measure the rate at which each sector sheds workers from its automatable, declining occupations, and compare it decade by decade with the recovered $\omega_t^{l,j}(\beta_t^j)\dot{\beta}_t^j$.

I use harmonized occupation counts from IPUMS-USA (Ruggles et al., 2024), pooling the decennial censuses over 1950–2000 with the American Community Survey over 2001–2019. I keep employed persons aged 16 and over, retain the self-employed, and assign each worker to a sector from the harmonized 1950 industry classification. Occupations are coded on the harmonized 1990 basis of 389 categories. Within a sector j and decade $[t_0, t_1]$, an occupation o is displaced if (i) its tasks are automatable by the technology available in that decade and (ii) its head count declines, $E_{o,t_1}^j < E_{o,t_0}^j$. I implement criterion (i) with Anthropic’s Claude Fable 5, a large language model (LLM), in two steps. First, for each occupation the LLM writes a short description of its core tasks. Second, it assigns the occupation a task category and the onset decade in which those tasks first become automatable, or “never” if they do not become automatable within my sample window.

Table 2 shows representative classifications.

Table 2: Representative entries from the occupation classification

Occupation	Task description	Task category	Automation onset
Telephone operators	Connect calls at switchboards	Routine clerical	1950–60
Farm workers	Manual farm field labor	Manual / physical	1950–60
Stenographers	Take dictation; transcribe	Routine clerical	1980–90
Bank tellers	Process deposits and withdrawals	Routine clerical	1980–90
Ticket and reservation agents	Book reservations; issue tickets	Routine clerical	2000–10
Registered nurses	In-person nursing care	Non-routine service	never
Accountants and auditors	Examine and audit accounts	Non-routine cognitive	never

Notes: Occupations are the 389 categories of the IPUMS-USA harmonized 1990 classification; the seven rows shown are selected to span the task categories and onset decades. Each occupation receives an LLM-written task description and an LLM-assigned task category and automation-onset decade; classifications were generated with Claude Fable 5 (Anthropic), with prompts and outputs recorded in the replication package. “Never” denotes not automatable within the 1950–2019 sample window, so occupations whose tasks became automatable only after 2019 (for example, through generative AI) are coded “never” for the purposes of this exercise.

The labor-side displacement rate is the annualized share of base-period sector employment shed from these occupations,

$$d_t^j = -\frac{1}{t_1 - t_0} \sum_{o \in D_t^j} \frac{E_{o,t_1}^j - E_{o,t_0}^j}{E_{t_0}^j}, \quad D_t^j = \{o : o \text{ automatable by } t_0 \text{ and } E_{o,t_1}^j < E_{o,t_0}^j\}.$$

The full occupation classification and construction are reported in the replication materials.

I read this measure as a lower bound on $\omega_t^{l,j}(\beta_t^j)\dot{\beta}_t^j$ in levels. Head counts register only workers who leave an occupation entirely, whereas much automation displaces tasks within an occupation, so the job persists while shedding tasks and hours to capital. However, the two measures should still be positively correlated.

Figure 2 plots the two displacement series by sector. The independent occupation measure co-moves with the recovered $\omega_t^{l,j}(\beta_t^j)\dot{\beta}_t^j$ at a correlation of 0.72 across the 21 sector–decade cells; as anticipated, it lies below $\omega_t^{l,j}(\beta_t^j)\dot{\beta}_t^j$ in levels. Repeating the construction on the harmonized 1950 and 2010 bases yields correlations of 0.71 and 0.61, and dropping the automatability gate altogether and computing d_t^j over all declining occupations yields 0.67.

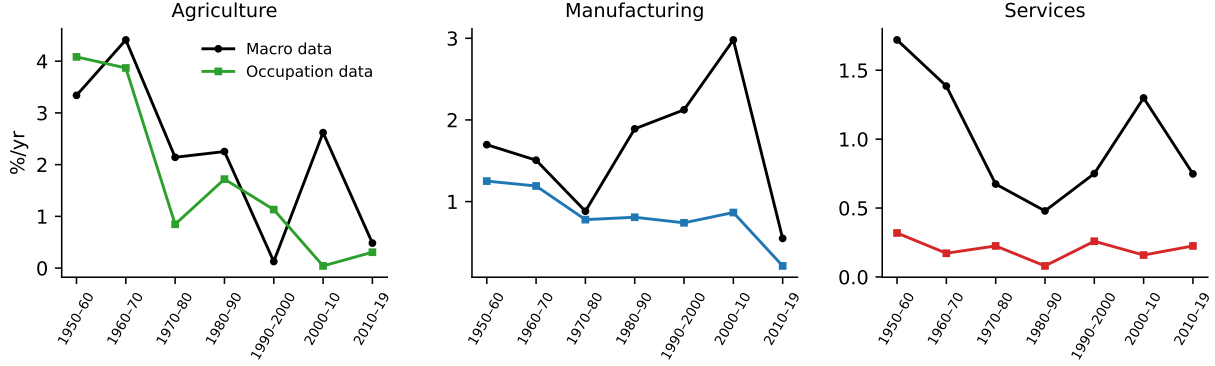


Figure 2: Recovered task-level displacement rate from macro data and the occupation-based displacement rate, by sector and decade.

5 Accounting for Employment Dynamics

With the technology series in hand, I now ask how much of the observed employment reallocation each channel accounts for. For each sector pair (j, m) , with manufacturing as the reference, averaging equation (16) over a period yields

$$\overline{\hat{L}^j} - \overline{\hat{L}^m} = \underbrace{\left(1 - \sigma + \frac{\rho}{1-\rho}\right)(\overline{\hat{P}^j} - \overline{\hat{P}^m})}_{\text{Price effect (PE)}} + \underbrace{(1 - \sigma)(\varepsilon_j - \varepsilon_m) \overline{\hat{c}_t}}_{\text{Income effect (IE)}} - \underbrace{(\omega^{l,j}(\beta^j)\dot{\beta}^j - \omega^{l,m}(\beta^m)\dot{\beta}^m)}_{\text{Automation's displacement effect (AE)}} + \underbrace{\overline{\xi_t^{j,m}}}_{\text{Residual}}, \quad (29)$$

where overlines denote period averages. The displacement effect's contribution is taken directly from Table 1. The price effect uses observed sectoral deflators. The income effect uses the growth rate of aggregate real consumption $\hat{c}_t$ constructed following [Comin et al. \(2021\)](#):

$$\hat{c}_t = \hat{W}_t - \hat{P}_t^m + \frac{1}{1 - \sigma} \hat{\omega}_t^m, \quad (30)$$

where $\hat{\omega}_t^m$ is the manufacturing expenditure share growth rate and W_t replaces the CLM nominal expenditure under the static budget constraint $\sum_j P_t^j c_t^j = W_t$. In the model the value-added shares coincide with consumption expenditure shares, so ω_t^m is measured directly from sectoral VA data, and per-person nominal expenditure W_t is constructed from the PWT. The parameters are calibrated as follows. From the demand side, $\sigma = 0.33$ and $(\varepsilon_a, \varepsilon_m, \varepsilon_s) = (0.01, 1, 1.94)$ are taken from [Comin et al. \(2021, Table X, col. 6\)](#) (value-added household estimation, CEX). From the supply side, $\rho = -2$ is taken from [Gechert et al. \(2022\)](#) as in Section 4.2. These deliver: the price-effect coefficient $1 - \sigma + \rho/(1 - \rho) = 0.003$, the income-effect coefficient for Agri-Mfg $(1 - \sigma)(\varepsilon_a - \varepsilon_m) = -0.66$, and the income-effect

coefficient for Svc-Mfg $(1 - \sigma)(\varepsilon_s - \varepsilon_m) = 0.63$. The price effect is thus negligible by calibration. Writing $\sigma_{KL} \equiv 1/(1 - \rho)$ for the elasticity of substitution across tasks, the employment price-effect coefficient is

$$1 - \sigma + \frac{\rho}{1 - \rho} = \sigma_{KL} - \sigma, \quad (31)$$

the gap between the supply-side and demand-side substitution elasticities: a rise in a sector's relative price raises labor demand per unit of output at rate σ_{KL} while lowering quantity demanded at rate σ , and employment responds to the difference. The coefficient is near zero under my baseline calibration because two independently estimated elasticities happen to nearly coincide.

Table 3 reports each channel as a share of the model-predicted relative employment reallocation in the corresponding sector pair and subperiod; the Explained column reports the model-predicted total as a share of the observed relative employment growth. The three model channels together explain 66–121% of observed cross-sector employment reallocation, depending on subperiod and sector pair.

Table 3: Decomposition of relative employment dynamics, US 1951–2019 ($\rho = -2$)

	Agri-Mfg				Svc-Mfg			
	PE	IE	AE	Explained	PE	IE	AE	Explained
1951–1989	0.24%	36.20%	63.56%	88.66%	0.16%	65.25%	34.59%	72.94%
1990–2019	−0.75%	−87.00%	187.74%	120.78%	0.25%	26.41%	73.35%	65.64%

Notes: PE, IE, AE are each channel's share of the model-predicted reallocation ($PE + IE + AE = 100\%$ by construction). Explained is the share of observed $\hat{L}^j - \hat{L}^m$ accounted for by the three channels jointly. Period averages over annual data.

In the pre-1990 period, for Agri-Mfg, AE accounts for 63.56% of the model-predicted reallocation against 36.20% for IE. For Svc-Mfg, IE leads at 65.25% while AE contributes 34.59%. In the post-1990 period, for the Agri-Mfg pair, the displacement channel rises further to 187.7% and the income channel turns negative (−87.0%). The sign flip on IE and PE arises because post-1990, IE and PE channels would predict agriculture's employment share to decline faster than manufacturing's, but in the data manufacturing's share falls faster. Only displacement pushes in the observed direction, and it is large enough to outweigh the other two channels. This directly follows the reordering documented in Section 4.2: manufacturing overtakes agriculture as the displacement leader post-1990. In the Svc-Mfg pair, the displacement channel rises to 73.3% of the model-predicted reallocation while the income channel falls to 26.4%, reflecting both the slowdown of

consumption-aggregator growth and the strengthening of the automation-displacement gap between manufacturing and services.

Figure 3 attributes the post-1950 evolution of the sectoral employment shares to the three channels of equation (29). I shut down the channels cumulatively (first the displacement effect, then the income effect, then the price effect), holding the remaining channels at their observed values; each counterfactual relative-employment path is integrated forward from its observed 1950 value and mapped back to shares.⁴ The gap between consecutive paths is a channel's shaded band in the figure.

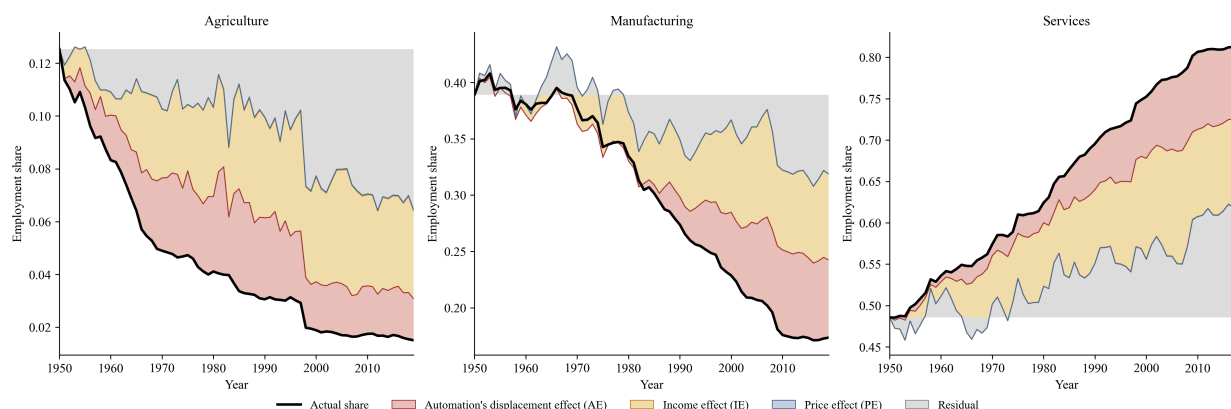


Figure 3: Decomposition of post-1950 employment-share changes into displacement, income, and price contributions, US 1950–2019 ($\rho = -2$). Solid black: actual shares. Shaded bands: cumulative direct contributions of automation's displacement effect (AE), the income effect (IE), the price effect (PE), and a residual.

The exercise quantifies how much of the realized cross-sector reallocation in employment over the postwar period is attributable directly to each channel. Cumulating to 2019, agriculture's share falls 11.0 percentage points below its 1950 level, of which the displacement channel directly contributes 1.6 points and the income channel 3.3; manufacturing's share falls by 21.6 points, with 6.9 from displacement and 7.6 from income; services' share stands 32.6 points above its 1950 level, with 8.4 from displacement and 11.0 from income. The price band is barely visible in all three panels, as the price-effect coefficient is nearly zero. The residual band absorbs the remainder (-6.1 , -7.0 , and $+13.1$ points, respectively).

The downward step in agriculture's actual share around 1998 reflects a change in industry classification (NIPA's SIC-to-NAICS transition). Several agricultural-services activities

⁴This partial-equilibrium construction isolates the direct contribution of each channel. I deliberately do not let the indirect, long-run feedback of cumulative automation through prices, wages, and the consumption aggregator operate, because the object of interest is the magnitude of each direct channel, in particular the displacement channel relative to the price and income effects, not the total general-equilibrium effect of automation.

(landscaping, veterinary services, etc.) were reclassified into the broader services aggregate. Agriculture's displacement was rapid before 1990, averaging 3.03% per year over 1951–1989 during the era of farm mechanization, and slowed to 1.11% afterward (Table 1). In manufacturing, displacement was low through the 1970s and then strengthened markedly from the mid-to-late 1980s. The decade-average rate more than doubled, from 0.88% per year in the 1970s to 1.89% in the 1980s, and rose further to 2.12% in the 1990s and 2.98% in the 2000s (Table 1). This timing coincides closely with the widespread adoption of industrial robots in US manufacturing, together with programmable controllers and computerized process automation, which diffused rapidly from the late 1980s onward. Accordingly, displacement contributes little to manufacturing's employment share before the mid-1980s and steadily more thereafter (Figure 3). In both subperiods, services consistently absorbed the workers displaced from agriculture and manufacturing, though this absorption accelerated after 1980: services' displacement and income bands widen from the 1980s onward.

6 Unexplained vs Buera and Kaboski (2009)

Buera and Kaboski (2009) (BK hereafter) document a large divergence between sectoral value-added and employment shares that, within a homogeneous-wage Cobb–Douglas benchmark, requires labor distortions of large magnitude, given by $(VA_t^j/L_t^j)/(VA_t^a/L_t^a)$ (see Figure 1). This paper's model likewise assumes a homogeneous wage, yet the macro data imply heterogeneous sectoral wages, leaving an analogous wedge $\mathcal{U}_t^j/\mathcal{U}_t^a$.⁵ Figure 4 plots the BK required distortion alongside this paper's unexplained wedge at the baseline ρ .⁶

The unexplained wedge averages $\bar{\mathcal{U}}^m/\bar{\mathcal{U}}^a = 1.40$ and $\bar{\mathcal{U}}^s/\bar{\mathcal{U}}^a = 1.37$ over 1950–2019, visually below the BK required distortion. That $\mathcal{U}_t^j/\mathcal{U}_t^a > 1$ in both pairs throughout the sample reveals the familiar agricultural wage gap: systematically higher wages in non-agriculture than in agriculture. Standard explanations (see Lagakos, 2020, for a review) combine lower average human capital among agricultural workers, with rural-to-urban

⁵Adding a sector-specific wedge $\mathcal{U}_t^j$ to the labor first-order condition (Appendix A.3) gives $\frac{\mathcal{U}_t^j}{\mathcal{U}_t^a} = \left(\frac{P_t^j}{P_t^a}\right)^{1/(1-\rho)} \cdot \frac{\int_{\beta_t^j}^1 \psi^{l,j}(i)^{\rho/(1-\rho)} di}{\int_{\beta_t^a}^1 \psi^{l,a}(i)^{\rho/(1-\rho)} di} \cdot \frac{Y_t^j/L_t^j}{Y_t^a/L_t^a}$. Under a homogeneous economy-wide wage, $\mathcal{U}_t^j/\mathcal{U}_t^a = 1$.

⁶To enable a direct comparison with BK, I take agriculture as the reference sector. The BK required distortion shown in Figure 4 is recomputed using the data of this paper, so that the BK series and the unexplained wedge $\mathcal{U}_t^j/\mathcal{U}_t^a$ are directly comparable.

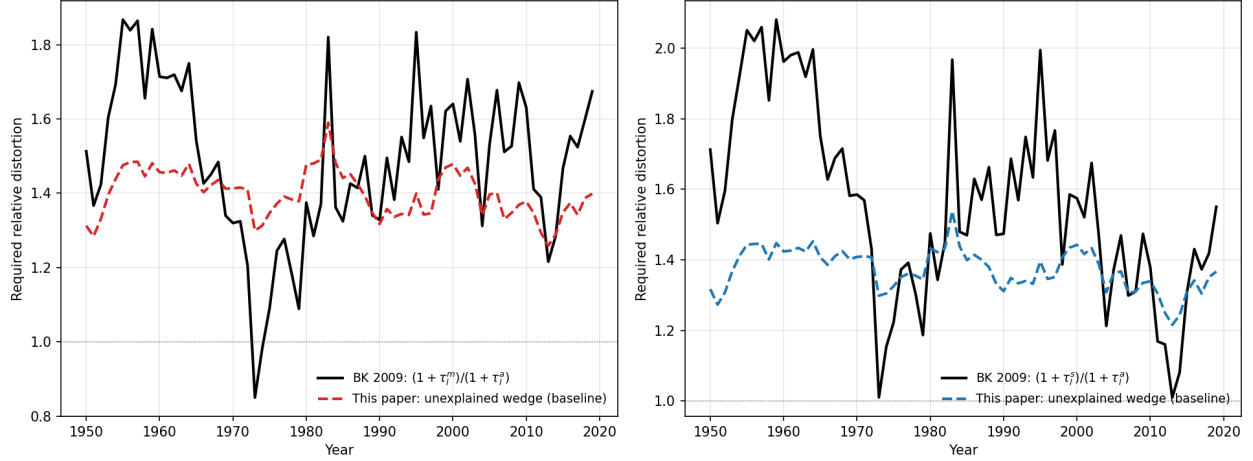


Figure 4: Buera–Kabuski (2009) required cross-sector distortion (solid) vs. this paper’s unexplained wedge $\mathcal{U}_t^j/\mathcal{U}_t^a$ (dashed), US 1950–2019 (baseline $\rho = -2$). Left: Manufacturing vs. Agriculture. Right: Services vs. Agriculture.

migration selecting on skill (Lagakos and Waugh, 2013; Herrendorf and Schoellman, 2018); a larger share of unpaid family labor and self-employment that biases measured productivity per worker downward (Herrendorf and Schoellman, 2015); and frictional mobility barriers such as credit constraints, location-specific human capital, and the loss of network-based insurance upon migration, which slow the arbitrage that would equalize wages across sectors (Munshi and Rosenzweig, 2016). The wedge narrows over 1950–2019, with decadal averages moving from roughly 1.40 in the 1950s to 1.32 in the 2010s, mirroring the long-run decline in the BK distortion since 1870 (Figure 1). The narrowing is consistent with a gradual erosion of the forces behind the gap: rising rural educational attainment shrinks the human-capital difference, and improved financial and geographic integration weakens the mobility barriers.

7 Robustness

The baseline results in Section 5 depend on two sets of calibrated parameters that are estimated outside this paper: the task substitution parameter ρ on the supply side, and preference parameters ($\sigma, \varepsilon_a, \varepsilon_m, \varepsilon_s$) on the demand side. This section examines the robustness of the baseline results along these two dimensions.

The baseline $\rho = -2$ ($\sigma_{KL} = 1/3$) matches the publication-bias-corrected best-practice estimate in Gechert et al. (2022). I try four additional values, $\rho \in \{-10, -3, -1, -0.5\}$. Table 4 reports automation’s displacement effect for each sector, averaged over the pre-1990 and post-1990 periods, at each ρ .

Table 4: Sectoral displacement rates $\omega^{l,j}(\beta^j)\dot{\beta}^j$ across ρ

ρ	σ_{KL}	Pre-1990			Post-1990		
		A	M	S	A	M	S
−10	0.09	3.93	2.01	1.40	1.87	2.35	1.22
−3	0.25	3.34	1.67	1.17	1.37	2.04	1.03
−2 (baseline)	0.33	3.03	1.49	1.05	1.11	1.88	0.94
−1	0.50	2.41	1.13	0.81	0.58	1.55	0.75
−0.5	0.67	1.79	0.77	0.56	0.06	1.23	0.56

Notes: Values are period averages in percent per year, by sector. A = Agriculture, M = Manufacturing, S = Services. Pre-1990: 1951–1989; Post-1990: 1990–2019.

Two patterns hold across every row of the table. First, pre-1990 the cross-sector ranking of displacement is $A > M > S$ for every ρ : agriculture is the unambiguous leader, manufacturing is second, services last. Second, post-1990 manufacturing overtakes agriculture and emerges as the cross-sector leader, with $M > A$ at every ρ and $M > S$ throughout. The reordering around 1990 is therefore robust. The post-1990 ordering between A and S does shift with ρ : at $\rho \in \{-10, -3, -2\}$ agriculture’s displacement remains above services, while at $\rho \in \{-1, -0.5\}$ services edges ahead.

Holding the demand-side parameters at CLM Table X col. 6 baseline values, Table 5 reports the decomposition of employment reallocation across ρ .

Table 5: Three-channel decomposition across ρ : shares of model-predicted reallocation

ρ (σ_{KL})	Agri–Mfg			Svc–Mfg		
	PE	IE	AE	PE	IE	AE
<i>1951–1989 (pre-1990)</i>						
−10 (0.09)	−17.6%	36.9%	80.6%	−11.0%	63.8%	47.2%
−3 (0.25)	−5.8%	36.5%	69.3%	−3.7%	64.8%	39.0%
−2, baseline (0.33)	0.2%	36.2%	63.6%	0.2%	65.2%	34.6%
−1 (0.50)	12.1%	35.7%	52.2%	8.1%	66.3%	25.6%
−0.5 (0.67)	23.5%	35.2%	41.2%	16.4%	67.3%	16.3%
<i>1990–2019 (post-1990)</i>						
−10 (0.09)	65.0%	−105.8%	140.8%	−18.1%	27.2%	90.9%
−3 (0.25)	19.1%	−92.7%	173.6%	−6.0%	26.7%	79.3%
−2, baseline (0.33)	−0.7%	−87.0%	187.7%	0.2%	26.4%	73.3%
−1 (0.50)	−33.9%	−77.5%	211.4%	12.3%	25.9%	61.8%
−0.5 (0.67)	−60.5%	−69.9%	230.4%	23.9%	25.4%	50.7%

Notes: Each cell reports the channel’s share of the model-predicted relative employment reallocation, computed from annual data at the baseline demand-side calibration.

Four patterns emerge. First, in three of the four cells (pre-1990 Agri-Mfg, pre-1990 Svc-Mfg, and post-1990 Svc-Mfg), the AE share is monotonically decreasing as $|\rho|$ shrinks from 10 to 0.5. The exception is post-1990 Agri-Mfg, where the AE share rises from 140.8% to 230.4%. Second, automation's displacement effect remains the dominant channel post-1990 in both sector pairs for every ρ . In Agri-Mfg the post-1990 AE share exceeds 140% throughout, with AE offsetting IE and PE, which push in the opposite direction. In Svc-Mfg the post-1990 AE share spans 50.7% to 90.9% and exceeds the IE and PE shares in every cell. Third, the sharp rise in the AE share between subperiods is preserved at every ρ . Fourth, the price channel's baseline negligibility is calibration-specific: its Agri-Mfg share moves from -17.6% at $\rho = -10$ to $+23.5\%$ at $\rho = -0.5$ pre-1990, flipping sign with $\sigma_{KL} - \sigma$. Displacement remains the largest channel post-1990 in both sector pairs at every ρ , displacement leads the pre-1990 Agri-Mfg cell at every ρ , and the income effect leads the pre-1990 Svc-Mfg cell at every ρ .

On the demand side, swapping the baseline parameters for CLM's final-good estimates, $(\sigma, \varepsilon_a, \varepsilon_m, \varepsilon_s) = (0.33, 0.03, 1.00, 1.85)$ from their Table I col. 6, moves every channel share by at most 2–3 percentage points at the baseline ρ (Appendix Table 11), because the estimated demand-side parameters differ little between CLM's two tables.

8 Conclusion

Why do sectoral value-added and employment shares often diverge during structural transformation? Are the two shares driven by different forces, and how quantitatively important is each? This paper builds a tractable model in which sectoral value-added dynamics are generated by price and income effects, while sectoral employment involves a third force, automation's displacement effect. I build a growth-accounting strategy on [Jones and Tonetti \(2026\)](#) to recover automation's displacement effect from standard macro data, and apply it to the US over 1950–2019. Automation's displacement effect accounts for 35–64% of model-predicted cross-sector employment reallocation before 1990 and 73–188% afterward, rising from a substantial channel to the largest of the three model channels. Automation's displacement and the income effect contribute comparably to the post-1950 changes in the employment shares, while the price effect contributes little at my baseline calibration.

The term $\omega_t^{l,j}(\beta_t^j)\dot{\beta}_t^j$ is measured directly from sectoral labor productivity and factor shares; its interpretation as automation's displacement requires the assumption of time-invariant task-level labor productivity. I discipline this assumption with evidence from outside the accounting system: the [Jones and Tonetti \(2026\)](#) estimates over samples beginning in 1950,

the [Acemoglu and Restrepo \(2022\)](#) task-displacement evidence for manufacturing, and an occupation-level validation correlating at 0.72 with the recovered series. Estimating $\psi^{l,j}(i)$ directly would require task-level data that do not yet exist; I leave this to future work.

My methodology travels: its inputs (real value added, employment, and the labor cost share by sector) are recorded in any KLEMS-style account, so the framework extends well beyond the US and three broad sectors. A non-US application needs, in addition, locally estimated demand and supply parameters; and the interpretive restriction $\hat{\psi}_t^{l,j} = 0$ must be reassessed economy by economy, most carefully where labor-augmenting agricultural change (such as seeds and fertiliser upgrade) is rapid. Whether employment is reallocating because automation displaces workers, because demand shifts away, or because distortions persist calls for different policy responses, and the decomposition delivers a first answer from data most economies already have. Looking ahead to AI, my decomposition cannot identify it separately from broader automation; but as adoption grows, the framework will reveal whether AI's displacement of labor is broad-based or concentrated in some sectors.

References

- Acemoglu, D. and Restrepo, P. (2018). The race between man and machine: Implications of technology for growth, factor shares, and employment. *American Economic Review*, 108(6):1488–1542.
- Acemoglu, D. and Restrepo, P. (2019). Automation and new tasks: How technology displaces and reinstates labor. *Journal of Economic Perspectives*, 33(2):3–30.
- Acemoglu, D. and Restrepo, P. (2020). Robots and jobs: Evidence from US labor markets. *Journal of Political Economy*, 128(6):2188–2244.
- Acemoglu, D. and Restrepo, P. (2022). Tasks, automation, and the rise in U.S. wage inequality. *Econometrica*, 90(5):1973–2016.
- Autor, D. H. and Dorn, D. (2013). The growth of low-skill service jobs and the polarization of the US labor market. *American Economic Review*, 103(5):1553–1597.
- Autor, D. H., Dorn, D., and Hanson, G. H. (2013). The china syndrome: Local labor market effects of import competition in the united states. *American Economic Review*, 103(6):2121–2168.
- Baumol, W. J. (1967). Macroeconomics of unbalanced growth: The anatomy of urban crisis. *American Economic Review*, 57(3):415–426.
- Buera, F. J. and Kaboski, J. P. (2009). Can traditional theories of structural change fit the data? *Journal of the European Economic Association*, 7(2–3):469–477.
- Comin, D., Lashkari, D., and Mestieri, M. (2021). Structural change with long-run income and price effects. *Econometrica*, 89(1):311–374.
- Feenstra, R. C., Inklaar, R., and Timmer, M. P. (2015). The next generation of the penn world table. *American Economic Review*, 105(10):3150–3182.
- Feenstra, R. C., Inklaar, R., and Timmer, M. P. (2025). Penn world table, version 11.0. Groningen Growth and Development Centre, University of Groningen.
- Gechert, S., Havranek, T., Irsova, Z., and Kolcunova, D. (2022). Measuring capital-labor substitution: The importance of method choices and publication bias. *Review of Economic Dynamics*, 45:55–82.
- Gollin, D. (2002). Getting income shares right. *Journal of Political Economy*, 110(2):458–474.
- Herrendorf, B., Herrington, C., and Valentinyi, Á. (2015). Sectoral technology and structural transformation. *American Economic Journal: Macroeconomics*, 7(4):104–133.

- Herrendorf, B., Rogerson, R., and Valentinyi, Á. (2013). Two perspectives on preferences and structural transformation. *American Economic Review*, 103(7):2752–2789.
- Herrendorf, B., Rogerson, R., and Valentinyi, Á. (2014). Growth and structural transformation. In Aghion, P. and Durlauf, S. N., editors, *Handbook of Economic Growth*, volume 2, chapter 6, pages 855–941. Elsevier, Amsterdam.
- Herrendorf, B. and Schoellman, T. (2015). Why is measured productivity so low in agriculture? *Review of Economic Dynamics*, 18(4):1003–1022.
- Herrendorf, B. and Schoellman, T. (2018). Wages, human capital, and barriers to structural transformation. *American Economic Journal: Macroeconomics*, 10(2):1–23.
- Jones, B. F. and Liu, X. (2024). A framework for economic growth with capital-embodied technical change. *American Economic Review*, 114(5):1448–1487.
- Jones, C. I. and Tonetti, C. (2026). Past automation and future AI: How weak links tame the growth explosion. Working Paper, Stanford GSB, Version 0.5, May 2026.
- Kongsamut, P., Rebelo, S., and Xie, D. (2001). Beyond balanced growth. *Review of Economic Studies*, 68(4):869–882.
- Lagakos, D. (2020). Urban-rural gaps in the developing world: Does internal migration offer opportunities? *Journal of Economic Perspectives*, 34(3):174–192.
- Lagakos, D. and Waugh, M. E. (2013). Selection, agriculture, and cross-country productivity differences. *American Economic Review*, 103(2):948–980.
- Manuelli, R. E. and Seshadri, A. (2014). Frictionless technology diffusion: The case of tractors. *American Economic Review*, 104(4):1368–1391.
- Munshi, K. and Rosenzweig, M. (2016). Networks and misallocation: Insurance, migration, and the rural-urban wage gap. *American Economic Review*, 106(1):46–98.
- Ngai, L. R. and Pissarides, C. A. (2007). Structural change in a multisector model of growth. *American Economic Review*, 97(1):429–443.
- Pierce, J. R. and Schott, P. K. (2016). The surprisingly swift decline of US manufacturing employment. *American Economic Review*, 106(7):1632–1662.
- Rodrik, D. (2016). Premature deindustrialization. *Journal of Economic Growth*, 21(1):1–33.
- Ruggles, S., Flood, S., Sobek, M., Backman, D., Chen, A., Cooper, G., Richards, S., Rodgers, R., and Schouweiler, M. (2024). IPUMS USA: Version 15.0.

Zeira, J. (1998). Workers, machines, and economic growth. *The Quarterly Journal of Economics*, 113(4):1091–1117.

A Proofs

A.1 Sectoral Prices and Growth Rate

Given the sectoral production function

$$Y_t^j = \left[\int_0^1 y_t^j(i)^\rho di \right]^{1/\rho}, \quad (\text{A1})$$

the dual price index is

$$P_t^j = \left[\int_0^1 p_t^j(i)^{-\frac{\rho}{1-\rho}} di \right]^{-\frac{1-\rho}{\rho}}. \quad (\text{A2})$$

Substituting the task-level unit costs

$$p_t^j(i) = \begin{cases} \frac{P_t^k}{\psi_t^{k,j}(i)}, & i \in [0, \beta_t^j], \\ \frac{W_t}{\psi_t^{l,j}(i)}, & i \in [\beta_t^j, 1], \end{cases} \quad (\text{A3})$$

gives the sectoral price aggregator

$$P_t^j = \left[\int_0^{\beta_t^j} \left(\frac{P_t^k}{\psi_t^{k,j}(i)} \right)^{-\frac{\rho}{1-\rho}} di + \int_{\beta_t^j}^1 \left(\frac{W_t}{\psi_t^{l,j}(i)} \right)^{-\frac{\rho}{1-\rho}} di \right]^{-\frac{1-\rho}{\rho}}. \quad (\text{A4})$$

Differentiating the price aggregator with respect to time,

$$\begin{aligned}
\dot{P}_t^j = & -\frac{1-\rho}{\rho} \left(P_t^j\right)^{\frac{\rho}{1-\rho}+1} \\
& \times \left\{ \int_0^{\beta_t^j} \left(\frac{1}{\psi_t^{k,j}(i)}\right)^{-\frac{\rho}{1-\rho}} \left(-\frac{\rho}{1-\rho}\right) \left(P_t^k\right)^{-\frac{\rho}{1-\rho}-1} \dot{P}_t^k \right. \\
& + \left(P_t^k\right)^{-\frac{\rho}{1-\rho}} \int_0^{\beta_t^j} \frac{\rho}{1-\rho} \psi_t^{k,j}(i)^{\frac{\rho}{1-\rho}-1} \dot{\psi}_t^{k,j}(i) di \\
& + \int_{\beta_t^j}^1 \left(\frac{1}{\psi_t^{l,j}(i)}\right)^{-\frac{\rho}{1-\rho}} di \left(-\frac{\rho}{1-\rho}\right) W_t^{-\frac{\rho}{1-\rho}-1} \dot{W}_t \\
& \left. + \underbrace{\left(\frac{P_t^k}{\psi_t^{k,j}(\beta_t^j)}\right)^{-\frac{\rho}{1-\rho}} \dot{\beta}_t^j - \left(\frac{W_t}{\psi_t^{l,j}(\beta_t^j)}\right)^{-\frac{\rho}{1-\rho}} \dot{\beta}_t^j}_{= 0 \text{ (break-even condition at } \beta_t^j)} \right\}. \tag{A5}
\end{aligned}$$

Converting to growth rates,

$$\begin{aligned}
\hat{P}_t^j = & \left(P_t^j\right)^{\frac{\rho}{1-\rho}} \left\{ \int_0^{\beta_t^j} \left(\frac{P_t^k}{\psi_t^{k,j}(i)}\right)^{-\frac{\rho}{1-\rho}} di \hat{P}_t^k \right. \\
& - \left(P_t^k\right)^{-\frac{\rho}{1-\rho}} \int_0^{\beta_t^j} \psi_t^{k,j}(i)^{\frac{\rho}{1-\rho}} \hat{\psi}_t^{k,j}(i) di \\
& \left. + \int_{\beta_t^j}^1 \left(\frac{W_t}{\psi_t^{l,j}(i)}\right)^{-\frac{\rho}{1-\rho}} di \hat{W}_t \right\}. \tag{A6}
\end{aligned}$$

Simplifying the expression using the sectoral factor share expressions in Appendix A.2 yields

$$\hat{P}_t^j = \alpha_t^j \hat{P}_t^k + (1 - \alpha_t^j) \hat{W}_t - \alpha_t^j \hat{\psi}_t^{k,j}. \tag{A7}$$

A.2 Factor Cost Shares and Growth Rate

Because the sectoral production function is CES across tasks, cost minimization yields the standard conditional input demands:

$$\psi_t^{k,j}(i) k_t^j(i) = \left(\frac{P_t^k / \psi_t^{k,j}(i)}{P_t^j} \right)^{-\frac{1}{1-\rho}} Y_t^j, \quad (\text{A8})$$

$$\psi_t^{l,j}(i) l_t^j(i) = \left(\frac{W_t / \psi_t^{l,j}(i)}{P_t^j} \right)^{-\frac{1}{1-\rho}} Y_t^j, \quad (\text{A9})$$

which can be rearranged as

$$k_t^j(i) = \left(P_t^k \right)^{-\frac{1}{1-\rho}} \psi_t^{k,j}(i)^{\frac{\rho}{1-\rho}} \left(P_t^j \right)^{\frac{1}{1-\rho}} Y_t^j, \quad (\text{A10})$$

$$l_t^j(i) = W_t^{-\frac{1}{1-\rho}} \psi_t^{l,j}(i)^{\frac{\rho}{1-\rho}} \left(P_t^j \right)^{\frac{1}{1-\rho}} Y_t^j. \quad (\text{A11})$$

Aggregating over the respective task ranges, the capital cost share is

$$\alpha_t^j = \frac{P_t^k K_t^j}{P_t^j Y_t^j} = \frac{P_t^k \int_0^{\beta_t^j} k_t^j(i) di}{P_t^j Y_t^j} = \left(P_t^k \right)^{-\frac{\rho}{1-\rho}} \left(P_t^j \right)^{\frac{\rho}{1-\rho}} \int_0^{\beta_t^j} \psi_t^{k,j}(i)^{\frac{\rho}{1-\rho}} di, \quad (\text{A12})$$

and the labor cost share is

$$1 - \alpha_t^j = \frac{W_t L_t^j}{P_t^j Y_t^j} = \frac{W_t \int_{\beta_t^j}^1 l_t^j(i) di}{P_t^j Y_t^j} = W_t^{-\frac{\rho}{1-\rho}} \left(P_t^j \right)^{\frac{\rho}{1-\rho}} \int_{\beta_t^j}^1 \psi_t^{l,j}(i)^{\frac{\rho}{1-\rho}} di. \quad (\text{A13})$$

Differentiating the cost shares with respect to time gives the capital cost share growth rate,

$$\hat{\alpha}_t^j = -\frac{\rho}{1-\rho} \left(1 - \alpha_t^j \right) \left(\hat{P}_t^k - \hat{W}_t - \hat{\psi}_t^{k,j} \right) + \omega_t^{k,j}(\beta_t^j) \dot{\beta}_t^j, \quad (\text{A14})$$

and the labor cost share growth rate,

$$\widehat{1 - \alpha_t^j} = -\frac{\rho}{1-\rho} \alpha_t^j \left(\hat{W}_t - \hat{P}_t^k + \hat{\psi}_t^{k,j} \right) - \omega_t^{l,j}(\beta_t^j) \dot{\beta}_t^j. \quad (\text{A15})$$

The levels of the cost shares depend on the position of the automation frontier but not on its movement; the frontier's movement enters only their growth rates.

A.3 Proof of Proposition 1

Value-added dynamics. This derivation follows [Comin et al. \(2021\)](#). The consumer's expenditure minimization problem is

$$\begin{aligned} \min_{c_t^j, j \in \{a,m,s\}} \quad & \sum_{j \in \{a,m,s\}} P_t^j c_t^j \\ \text{s.t.} \quad & \sum_{j \in \{a,m,s\}} \Omega_j^{\frac{1}{\sigma}} \left(\frac{c_t^j}{\varepsilon_j} \right)^{\frac{\sigma-1}{\sigma}} = 1. \end{aligned} \quad (\text{A16})$$

The Lagrangian is

$$\mathcal{L} = \sum_{j \in \{a,m,s\}} P_t^j c_t^j + \lambda_t \left[1 - \sum_{j \in \{a,m,s\}} \Omega_j^{\frac{1}{\sigma}} \left(\frac{c_t^j}{\varepsilon_j} \right)^{\frac{\sigma-1}{\sigma}} \right]. \quad (\text{A17})$$

The first-order condition with respect to c_t^j is

$$P_t^j = \lambda_t \frac{\sigma-1}{\sigma} \Omega_j^{\frac{1}{\sigma}} c_t^{-\varepsilon_j \frac{\sigma-1}{\sigma}} (c_t^j)^{-\frac{1}{\sigma}}. \quad (\text{A18})$$

Solving for c_t^j :

$$c_t^j = (P_t^j)^{-\sigma} \left[\frac{\sigma-1}{\sigma} \lambda_t \Omega_j^{\frac{1}{\sigma}} c_t^{-\varepsilon_j \frac{\sigma-1}{\sigma}} \right]^{\sigma}. \quad (\text{A19})$$

Substituting into the constraint and solving for λ_t :

$$\lambda_t = \frac{\sigma}{\sigma-1} \left(\sum_{j \in \{a,m,s\}} \Omega_j (P_t^j)^{1-\sigma} c_t^{\varepsilon_j(1-\sigma)} \right)^{\frac{1}{1-\sigma}}. \quad (\text{A20})$$

Substituting back into the expression for c_t^j :

$$c_t^j = (P_t^j)^{-\sigma} \left(\sum_{j \in \{a,m,s\}} \Omega_j (P_t^j)^{1-\sigma} c_t^{\varepsilon_j(1-\sigma)} \right)^{\frac{\sigma}{1-\sigma}} \Omega_j^{\frac{1}{\sigma}} c_t^{\varepsilon_j(1-\sigma)}. \quad (\text{A21})$$

Total expenditure is therefore

$$E_t = \sum_{j \in \{a,m,s\}} P_t^j c_t^j = \left(\sum_{j \in \{a,m,s\}} \Omega_j (P_t^j)^{1-\sigma} c_t^{\varepsilon_j(1-\sigma)} \right)^{\frac{1}{1-\sigma}}, \quad (\text{A22})$$

which gives the Marshallian demand

$$c_t^j = \Omega_j \left(\frac{P_t^j}{E_t} \right)^{-\sigma} c_t^{(1-\sigma)\varepsilon_j}. \quad (\text{A23})$$

Aggregate sectoral value added is $VA_t^j = L_t P_t^j c_t^j$ by goods market clearing. Taking growth rates and differencing across sectors j and j' , population growth $\hat{L}_t$ cancels:

$$\widehat{VA}_t^j - \widehat{VA}_t^{j'} = (1-\sigma) (\hat{P}_t^j - \hat{P}_t^{j'}) + (1-\sigma)(\varepsilon_j - \varepsilon_{j'}) \hat{c}_t, \quad (\text{A24})$$

which is equation (15).

Employment dynamics. From the task-level labor demand derived in Appendix A.2, sectoral labor demand aggregates to

$$L_t^j = W_t^{-\frac{1}{1-\rho}} \int_{\beta_t^j}^1 \psi^{l,j}(i)^{\frac{\rho}{1-\rho}} di \left(P_t^j \right)^{\frac{1}{1-\rho}} Y_t^j. \quad (\text{A25})$$

With $\psi^{l,j}(i)$ time-invariant, the Leibniz rule and (13) give the growth rate of the integral as $-\omega_t^{l,j}(\beta_t^j) \dot{\beta}_t^j$. Taking growth rates and differencing across sectors j and j' , the wage terms cancel; substituting $\hat{Y}_t^j - \hat{Y}_t^{j'} = -\sigma(\hat{P}_t^j - \hat{P}_t^{j'}) + (1-\sigma)(\varepsilon_j - \varepsilon_{j'}) \hat{c}_t$, which follows from goods-market clearing and the Marshallian demand (3), yields

$$\hat{L}_t^j - \hat{L}_t^{j'} = \left(1 - \sigma + \frac{\rho}{1-\rho} \right) (\hat{P}_t^j - \hat{P}_t^{j'}) + (1-\sigma)(\varepsilon_j - \varepsilon_{j'}) \hat{c}_t - \left(\omega_t^{l,j}(\beta_t^j) \dot{\beta}_t^j - \omega_t^{l,j'}(\beta_t^{j'}) \dot{\beta}_t^{j'} \right), \quad (\text{A26})$$

which is (16).

A.4 Proof of Lemma 1

Take the wage as numeraire and write $p_t^j \equiv P_t^j/W_t$ for $j \in \{k, a, m, s\}$. In relative form, the sectoral price index (11) defines the unit-cost function

$$p_t^j = C^j(p_t^k; \beta_t^j) \equiv \left[\int_0^{\beta_t^j} \left(\frac{p_t^k}{\psi_t^{k,j}(i)} \right)^{-\frac{\rho}{1-\rho}} di + \int_{\beta_t^j}^1 \left(\frac{1}{\psi_t^{l,j}(i)} \right)^{-\frac{\rho}{1-\rho}} di \right]^{-\frac{1-\rho}{\rho}}. \quad (\text{A27})$$

The task-level productivity profiles $\psi_t^{k,j}(\cdot)$ and $\psi_t^{l,j}(\cdot)$ enter C^j as fixed parameters: the lemma perturbs the automation frontier alone, holding them constant. The proof proceeds in four steps: the automation frontier does not move a sector's own unit cost (Step 1); it therefore moves neither the capital-good price (Step 2) nor any sectoral price (Step 3); and prices are the only channel into consumption (Step 4).

Step 1: the automation frontier does not move the unit cost. Holding p_t^k fixed, C^j depends on β_t^j only through the integration boundary, and the Leibniz rule gives

$$\frac{\partial C^j}{\partial \beta_t^j} \propto \left(\frac{p_t^k}{\psi_t^{k,j}(\beta_t^j)} \right)^{-\frac{\rho}{1-\rho}} - \left(\frac{1}{\psi_t^{l,j}(\beta_t^j)} \right)^{-\frac{\rho}{1-\rho}} = 0, \quad (\text{A28})$$

by the break-even condition (8): the marginal task costs the same under either factor, so reassigning it has no first-order effect on the minimized cost. Differentiating (A27) with respect to p_t^k instead gives Shephard's lemma,

$$\frac{\partial \ln C^j}{\partial \ln p_t^k} = \frac{\int_0^{\beta_t^j} (p_t^k / \psi_t^{k,j}(i))^{-\frac{\rho}{1-\rho}} di}{\int_0^{\beta_t^j} (p_t^k / \psi_t^{k,j}(i))^{-\frac{\rho}{1-\rho}} di + \int_{\beta_t^j}^1 (1 / \psi_t^{l,j}(i))^{-\frac{\rho}{1-\rho}} di} = \alpha_t^j \in (0, 1), \quad (\text{A29})$$

with $\alpha_t^j < 1$ because $\beta_t^j < 1$ leaves a positive measure of labor tasks.

Step 2: the capital-good price does not move. Competition prices sector k 's output at its unit cost, so p_t^k solves the fixed point $p_t^k = C^k(p_t^k; \beta_t^k)$, an equation in which no other sector's automation frontier appears. Differentiating in logs with respect to β_t^k , the boundary term vanishes by (A28); what remains is the feedback of the capital-good price on its own unit cost, which by (A29) reads $d \ln p_t^k / d \beta_t^k = \alpha_t^k d \ln p_t^k / d \beta_t^k$. Since $\alpha_t^k < 1$, $d \ln p_t^k / d \beta_t^k = 0$. The capital-good price is therefore invariant to every automation frontier.

Step 3: no sectoral price moves. For $j \in \{a, m, s\}$, the unit cost $C^j(p_t^k; \beta_t^j)$ involves only the sector's own automation frontier and the capital-good price. The own-frontier effect vanishes by Step 1, and the p_t^k channel is shut by Step 2, so no automation frontier moves any sectoral price.

Step 4: consumption does not move. Substituting the sectoral expenditure (4) into the budget constraint $\sum_{j \in \{a, m, s\}} P_t^j c_t^j = W_t$ and dividing by W_t gives the implicit equation for c_t ,

$$F(c_t; p_t^a, p_t^m, p_t^s) \equiv \sum_{j \in \{a, m, s\}} \Omega_j \left(p_t^j \right)^{1-\sigma} c_t^{(1-\sigma)\varepsilon_j} - 1 = 0. \quad (\text{A30})$$

The automation frontiers enter F only through the relative prices. With $\sigma \in (0, 1)$ and $\varepsilon_j > 0$, F is strictly increasing in c_t , so the implicit function theorem applies, and $dc_t/d\beta_t^j = 0$ by Step 3. $\square$

A.5 CES Representation of the Task-Based Production Function

The task-based production function admits a two-factor CES representation at the sectoral level. Starting from

$$\begin{aligned} Y_t^j &= \left[\int_0^1 y_t^j(i)^\rho di \right]^{1/\rho} \\ &= \left[\int_0^{\beta_t^j} \left(\psi_t^{k,j}(i) k_t^j(i) \right)^\rho di + \int_{\beta_t^j}^1 \left(\psi_t^{l,j}(i) l_t^j(i) \right)^\rho di \right]^{1/\rho}, \end{aligned} \quad (\text{A31})$$

Factoring $(K_t^j)^\rho$ and $(L_t^j)^\rho$ out of the two task blocks and imposing the cost-minimizing within-factor allocations

$$\frac{k_t^j(i)}{K_t^j} = \frac{\psi_t^{k,j}(i)^{\frac{\rho}{1-\rho}}}{\int_0^{\beta_t^j} \psi_t^{k,j}(i')^{\frac{\rho}{1-\rho}} di'}, \quad \frac{l_t^j(i)}{L_t^j} = \frac{\psi_t^{l,j}(i)^{\frac{\rho}{1-\rho}}}{\int_{\beta_t^j}^1 \psi_t^{l,j}(i')^{\frac{\rho}{1-\rho}} di'}, \quad (\text{A32})$$

to obtain

$$\begin{aligned} Y_t^j &= \left[\left(\int_0^{\beta_t^j} \psi_t^{k,j}(i)^{\frac{\rho}{1-\rho}} di \right)^{1-\rho} (K_t^j)^\rho + \left(\int_{\beta_t^j}^1 \psi_t^{l,j}(i)^{\frac{\rho}{1-\rho}} di \right)^{1-\rho} (L_t^j)^\rho \right]^{1/\rho} \\ &= \left[(B_t^j K_t^j)^\rho + (A_t^j L_t^j)^\rho \right]^{1/\rho}, \end{aligned} \quad (\text{A33})$$

where the composite technology indices are

$$B_t^j = \left(\int_0^{\beta_t^j} \psi_t^{k,j}(i)^{\frac{\rho}{1-\rho}} di \right)^{\frac{1-\rho}{\rho}}, \quad A_t^j = \left(\int_{\beta_t^j}^1 \psi_t^{l,j}(i)^{\frac{\rho}{1-\rho}} di \right)^{\frac{1-\rho}{\rho}}. \quad (\text{A34})$$

A.6 Measurement of A_t^j

From the two-factor CES representation of Appendix A.5, cost minimization equates labor's value marginal product to its price, so the labor cost share is the CES value share

$$1 - \alpha_t^j = \frac{W_t L_t^j}{P_t^j Y_t^j} = \frac{\partial Y_t^j}{\partial L_t^j} \frac{L_t^j}{Y_t^j} = \frac{(A_t^j L_t^j)^\rho}{(Y_t^j)^\rho}. \quad (\text{A35})$$

Rearranging gives

$$A_t^j = (1 - \alpha_t^j)^{1/\rho} \frac{Y_t^j}{L_t^j}. \quad (\text{A36})$$

A.7 TFP Accounting Identity

By the argument of Appendix A.6, applied to both factors, the cost shares are the CES value shares

$$\alpha_t^j = \frac{P_t^k K_t^j}{P_t^j Y_t^j} = \frac{\partial Y_t^j}{\partial K_t^j} \frac{K_t^j}{Y_t^j} = \frac{(B_t^j K_t^j)^\rho}{(Y_t^j)^\rho}, \quad 1 - \alpha_t^j = \frac{(A_t^j L_t^j)^\rho}{(Y_t^j)^\rho}. \quad (\text{A37})$$

Differentiating the production function $(Y_t^j)^\rho = (B_t^j K_t^j)^\rho + (A_t^j L_t^j)^\rho$ and using these shares, output growth is

$$\begin{aligned}\hat{Y}_t^j &= \frac{(B_t^j K_t^j)^\rho (\hat{B}_t^j + \hat{K}_t^j) + (A_t^j L_t^j)^\rho (\hat{A}_t^j + \hat{L}_t^j)}{(Y_t^j)^\rho} \\ &= \alpha_t^j (\hat{B}_t^j + \hat{K}_t^j) + (1 - \alpha_t^j) (\hat{A}_t^j + \hat{L}_t^j).\end{aligned}\quad (\text{A38})$$

Total factor productivity growth is the Solow residual, output growth net of the cost-share-weighted input growth,

$$\widehat{\text{TFP}}_t^j = \hat{Y}_t^j - \alpha_t^j \hat{K}_t^j - (1 - \alpha_t^j) \hat{L}_t^j = \alpha_t^j \hat{B}_t^j + (1 - \alpha_t^j) \hat{A}_t^j. \quad (\text{A39})$$

A.8 Task-Based Mappings

The three mappings (24)–(26) follow from differentiating the productivity aggregates (18)–(19) and imposing the break-even condition (8).

The labor aggregate (19) depends on time only through its lower limit β_t^j , so differentiating it gives $\hat{A}_t^j = -\frac{1-\rho}{\rho} \omega_t^{l,j}(\beta_t^j) \dot{\beta}_t^j$. Rearranging gives (25),

$$\omega_t^{l,j}(\beta_t^j) \dot{\beta}_t^j = -\frac{\rho}{1-\rho} \hat{A}_t^j. \quad (\text{A40})$$

At the marginal task, the break-even condition (8) equates the unit cost of capital and labor production, $P_t^k / \psi_t^{k,j}(\beta_t^j) = W_t / \psi_t^{l,j}(\beta_t^j)$, indicating equal expenditure under either factor, $P_t^k k_t^j(\beta_t^j) = W_t l_t^j(\beta_t^j)$. Dividing by $P_t^j Y_t^j$, with $\alpha_t^j \omega_t^{k,j}(\beta_t^j) = P_t^k k_t^j(\beta_t^j) / (P_t^j Y_t^j)$ and $(1 - \alpha_t^j) \omega_t^{l,j}(\beta_t^j) = W_t l_t^j(\beta_t^j) / (P_t^j Y_t^j)$, gives the equality

$$\alpha_t^j \omega_t^{k,j}(\beta_t^j) = (1 - \alpha_t^j) \omega_t^{l,j}(\beta_t^j). \quad (\text{A41})$$

Multiplying this equality by $\dot{\beta}_t^j$ and substituting (25) gives (24),

$$\omega_t^{k,j}(\beta_t^j) \dot{\beta}_t^j = \frac{1 - \alpha_t^j}{\alpha_t^j} \omega_t^{l,j}(\beta_t^j) \dot{\beta}_t^j = -\frac{\rho}{1-\rho} \frac{1 - \alpha_t^j}{\alpha_t^j} \hat{A}_t^j. \quad (\text{A42})$$

Differentiating the capital aggregate (18) gives $\hat{B}_t^j = \hat{\psi}_t^{k,j} + \frac{1-\rho}{\rho} \omega_t^{k,j}(\beta_t^j) \dot{\beta}_t^j$. The cost-share-

weighted combination of the two growth rates is then

$$\begin{aligned}
\widehat{\text{TFP}}_t^j &= \alpha_t^j \hat{B}_t^j + (1 - \alpha_t^j) \hat{A}_t^j \\
&= \alpha_t^j \hat{\psi}_t^{k,j} + \frac{1 - \rho}{\rho} \left(\alpha_t^j \omega_t^{k,j}(\beta_t^j) - (1 - \alpha_t^j) \omega_t^{l,j}(\beta_t^j) \right) \hat{\beta}_t^j \\
&= \alpha_t^j \hat{\psi}_t^{k,j},
\end{aligned} \tag{A43}$$

Dividing by α_t^j gives (26),

$$\hat{\psi}_t^{k,j} = \hat{B}_t^j + \frac{1 - \alpha_t^j}{\alpha_t^j} \hat{A}_t^j. \tag{A44}$$

B Data Sources

Table 6 lists the primary sources for each series used in the empirical analysis of Sections 4 and 5, split into two time spans due to the SIC-NAICS classification change. Per-capita consumption expenditure and the rent paid to nonoperator landlords are common to both periods. Construction details are documented in the replication package accompanying the paper.

Table 6: Data sources

Series	1950–1997	1998–2022
Value added (nominal and real)	BEA GDP-by-Industry (1947–97 NAICS-basis historical series)	BEA GDP-by-Industry (NAICS)
Employment	NIPA Table 6.8 Persons Engaged (SIC)	NIPA Table 6.8 Persons Engaged (NAICS)
Factor share	BEA SIC GDP-by-Industry	BEA NAICS components; PI from NIPA Tables 6.12 and 1.13
TFP	BEA–BLS Integrated ILPA (El-dridge)	BEA–BLS Integrated ILPA
Per person expenditure	Penn World Table 11.0 (Feenstra et al., 2025, 2015)	
Rent paid to nonoperator landlords	NIPA Table 7.3.5 and 7.3.3	

C Sensitivity Analysis

This appendix reports three complements to the robustness exercise of Section 7: (i) the underlying sector-level technology series at each non-baseline ρ (Tables 7–10), (ii) the counterfactual employment-share trajectories with the displacement channel shut down at

each non-baseline ρ (Figure 5), and (iii) the decomposition under both of CLM’s demand-side calibrations (Table 11).

C.1 Technology Measurement at Non-Baseline ρ

This subsection reports three sector-level technology series for the four non-baseline values of ρ in the robustness range, namely $\rho \in \{-10, -3, -1, -0.5\}$: $\omega_t^{k,j}(\beta_t^j)\dot{\beta}_t^j$ (frontier contribution to capital deepening), $\omega_t^{l,j}(\beta_t^j)\dot{\beta}_t^j$ (frontier contribution to labor displacement), and $\hat{\psi}_t^{k,j}$ (capital-augmenting productivity growth at already-automated tasks). The baseline $\rho = -2$ table appears in Section 4.2 (Table 1). $\hat{\psi}_t^{k,j}$ is ρ -invariant.

Table 7: Sectoral technology measurements ($\rho = -10, \sigma_{KL} = 0.09$)

	$\omega_t^{k,j}(\beta_t^j)\dot{\beta}_t^j$			$\omega_t^{l,j}(\beta_t^j)\dot{\beta}_t^j$			$\hat{\psi}_t^{k,j}$		
	A	M	S	A	M	S	A	M	S
1950s	3.06	6.81	3.55	4.05	2.48	2.20	5.75	3.12	1.64
1960s	3.56	5.98	3.11	5.48	2.16	2.00	6.03	7.06	2.50
1970s	1.25	3.27	1.61	2.36	1.13	0.94	2.48	-0.64	1.89
1980s	1.67	5.82	0.80	3.83	2.34	0.53	6.53	4.13	-0.48
1990s	0.80	5.99	1.75	1.10	2.71	1.08	6.25	4.55	0.51
2000s	2.02	5.65	2.55	3.90	3.45	1.58	5.08	5.03	0.72
2010s	0.31	1.15	1.41	0.55	0.92	0.99	-0.05	1.22	1.28
Pre-1990	2.37	5.44	2.23	3.93	2.01	1.40	5.19	3.43	1.38
Post-1990	1.05	4.20	1.91	1.87	2.35	1.22	3.68	3.57	0.85

Notes: Values in percent per year. A = Agriculture, M = Manufacturing, S = Services. Decade and sub-period entries are period averages of annual values. Pre-1990 = 1951–1989; Post-1990 = 1990–2019.

Table 8: Sectoral technology measurements ($\rho = -3, \sigma_{KL} = 0.25$)

	$\omega^{k,j}(\beta^j)\dot{\beta}^j$			$\omega^{l,j}(\beta^j)\dot{\beta}^j$			$\hat{\psi}^{k,j}$		
	A	M	S	A	M	S	A	M	S
1950s	2.71	5.41	3.04	3.58	1.97	1.89	5.75	3.12	1.64
1960s	3.10	4.79	2.48	4.78	1.73	1.60	6.03	7.06	2.50
1970s	1.17	2.81	1.31	2.22	0.97	0.77	2.48	-0.64	1.89
1980s	1.19	5.10	0.75	2.80	2.04	0.50	6.53	4.13	-0.48
1990s	0.38	5.14	1.40	0.46	2.32	0.86	6.25	4.55	0.51
2000s	1.58	5.11	2.25	3.06	3.14	1.39	5.08	5.03	0.72
2010s	0.28	0.84	1.19	0.51	0.68	0.83	-0.05	1.22	1.28
Pre-1990	2.03	4.50	1.87	3.34	1.67	1.17	5.19	3.43	1.38
Post-1990	0.76	3.65	1.62	1.37	2.04	1.03	3.68	3.57	0.85

Notes: Values in percent per year. A = Agriculture, M = Manufacturing, S = Services. Decade and sub-period entries are period averages of annual values. Pre-1990 = 1951–1989; Post-1990 = 1990–2019.

Table 9: Sectoral technology measurements ($\rho = -1, \sigma_{KL} = 0.50$)

	$\omega^{k,j}(\beta^j)\dot{\beta}^j$			$\omega^{l,j}(\beta^j)\dot{\beta}^j$			$\hat{\psi}^{k,j}$		
	A	M	S	A	M	S	A	M	S
1950s	2.18	3.21	2.24	2.85	1.16	1.39	5.75	3.12	1.64
1960s	2.38	2.91	1.49	3.68	1.06	0.96	6.03	7.06	2.50
1970s	1.04	2.08	0.84	1.99	0.71	0.49	2.48	-0.64	1.89
1980s	0.43	3.96	0.68	1.17	1.58	0.44	6.53	4.13	-0.48
1990s	-0.27	3.79	0.86	-0.54	1.72	0.53	6.25	4.55	0.51
2000s	0.90	4.27	1.78	1.74	2.66	1.11	5.08	5.03	0.72
2010s	0.24	0.35	0.83	0.44	0.29	0.58	-0.05	1.22	1.28
Pre-1990	1.49	3.04	1.29	2.41	1.13	0.81	5.19	3.43	1.38
Post-1990	0.31	2.77	1.17	0.58	1.55	0.75	3.68	3.57	0.85

Notes: Values in percent per year. A = Agriculture, M = Manufacturing, S = Services. Decade and sub-period entries are period averages of annual values. Pre-1990 = 1951–1989; Post-1990 = 1990–2019.

Table 10: Sectoral technology measurements ($\rho = -0.5$, $\sigma_{KL} = 0.67$)

	$\omega^{k,j}(\beta^j)\dot{\beta}^j$			$\omega^{l,j}(\beta^j)\dot{\beta}^j$			$\hat{\psi}^{k,j}$		
	A	M	S	A	M	S	A	M	S
1950s	1.82	1.74	1.70	2.36	0.62	1.06	5.75	3.12	1.64
1960s	1.90	1.66	0.83	2.94	0.61	0.54	6.03	7.06	2.50
1970s	0.95	1.59	0.53	1.85	0.55	0.31	2.48	-0.64	1.89
1980s	-0.08	3.21	0.64	0.08	1.28	0.41	6.53	4.13	-0.48
1990s	-0.71	2.90	0.49	-1.21	1.32	0.30	6.25	4.55	0.51
2000s	0.44	3.71	1.46	0.86	2.34	0.92	5.08	5.03	0.72
2010s	0.22	0.02	0.60	0.39	0.03	0.42	-0.05	1.22	1.28
Pre-1990	1.13	2.06	0.90	1.79	0.77	0.56	5.19	3.43	1.38
Post-1990	0.01	2.19	0.86	0.06	1.23	0.56	3.68	3.57	0.85

Notes: Values in percent per year. A = Agriculture, M = Manufacturing, S = Services. Decade and sub-period entries are period averages of annual values. Pre-1990 = 1951–1989; Post-1990 = 1990–2019.

C.2 Counterfactual Employment Shares at Non-Baseline ρ

Figure 5 reports the no-displacement counterfactual of Figure 3 (employment shares with the displacement channel shut down, holding the income and price channels at their observed values) for the five ρ values in the robustness range. The counterfactual gap is increasing in $|\rho|$, but the qualitative pattern is preserved throughout: without displacement, agriculture's and manufacturing's shares are substantially higher, and services' is substantially lower.

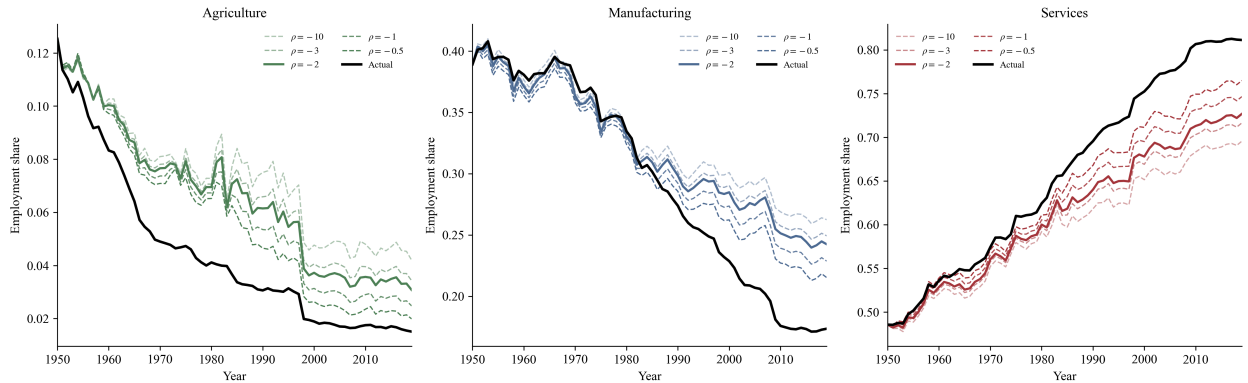


Figure 5: Counterfactual employment shares with displacement shut down: sensitivity to ρ , US 1950–2019. Solid black: actual shares. Dashed lines: counterfactual at $\rho \in \{-10, -3, -1, -0.5\}$; solid colored: counterfactual at baseline $\rho = -2$.

C.3 Demand-Side Parameters

Table 11 reports the three-channel decomposition under both of CLM’s demand-side calibrations, holding $\rho = -2$ fixed.

Table 11: Sensitivity of the decomposition to demand-side calibration ($\rho = -2$)

	Agri-Mfg				Svc-Mfg			
	PE	IE	AE	Explained	PE	IE	AE	Explained
<i>1951–1989 (pre-1990)</i>								
VA, baseline	0.24%	36.20%	63.56%	88.66%	0.16%	65.25%	34.59%	72.94%
Final-good	0.24%	35.73%	64.03%	88.01%	0.17%	62.94%	36.89%	68.38%
<i>1990–2019 (post-1990)</i>								
VA, baseline	−0.75%	−87.00%	187.74%	120.78%	0.25%	26.41%	73.35%	65.64%
Final-good	−0.73%	−83.77%	184.50%	122.90%	0.25%	24.50%	75.25%	63.98%

Notes: Both rows use demand-system estimates of [Comin et al. \(2021\)](#): the VA baseline from their Table X, $(\varepsilon_a, \varepsilon_m, \varepsilon_s) = (0.01, 1.00, 1.94)$, and the final-good specification from their Table I, $(\varepsilon_a, \varepsilon_m, \varepsilon_s) = (0.03, 1.00, 1.85)$.

D Accounting for Value-Added Dynamics

By construction, the model implies that relative value-added growth is driven by price and income effects:

$$\widehat{VA}_t^j - \widehat{VA}_t^m = \underbrace{(1 - \sigma)(\hat{P}_t^j - \hat{P}_t^m)}_{\text{PE}} + \underbrace{(1 - \sigma)(\varepsilon_j - \varepsilon_m)}_{\text{IE}} \hat{c}_t.$$

Table 12 reports the decomposition based on this equation, at the baseline calibration $(\sigma, \varepsilon_a, \varepsilon_m, \varepsilon_s) = (0.33, 0.01, 1.00, 1.94)$ from [Comin et al. \(2021, Table X col. 6\)](#), in the same period split as Table 3.

Table 12: Decomposition of relative value-added dynamics, US 1951–2019

	Agri-Mfg			Svc-Mfg		
	PE	IE	Explained	PE	IE	Explained
1951–1989	57.11%	42.89%	84.40%	32.59%	67.41%	73.99%
1990–2019	63.26%	36.74%	103.15%	65.14%	34.86%	57.38%

In Table 12 the price effect generally dominates the income effect, in line with [Herrendorf et al. \(2013\)](#), who find the same for postwar US consumption value added.